Multiple case assignment in ATB-movement: Evidence from non-syncretic mismatches in Russian

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Abstract

It is a well-established fact that ATB-movement is subject to a case-matching requirement. Exceptions are usually only found with syncretic forms. In this paper we show that certain non-syncretic mismatches are exceptionally possible with the genitive of negation. Based on new data from Russian, we will propose an account that involves case decomposition and multiple case assignment. We will argue that non-syncretic mismatches are possible under these assumptions because the genitive featurally contains the accusative (but not other cases). The data require an asymmetric approach to ATB-movement where the case in the second conjunct is assigned first, and we will present an account in terms of sideward movement (Nunes 2004). Our approach based on multiple case assignment also provides a new perspective on the distribution of genitive of negation within a single clause. Finally, we argue that the phenomenon at hand is instructive regarding the viability of different approaches to ATB-movement in that it is incompatible with most of them.
Keywords: ATB-movement, genitive of negation, case-matching, syncretism, multiple case assignment, case decomposition, Russian, sideward movement

1 Introduction

The goals of this paper are twofold. First, it investigates case, the mechanisms of case assignment, and case decomposition, and, second, it contributes to the ongoing discussion on the syntax of ATB-movement.

Beginning with the first issue, in current minimalist syntax, case assignment is typically treated as a unidirectional Agree relation between a probe and a goal, whereby a nominal receives a value for its initially unvalued case feature (Chomsky 2000, 2001). Once Agree has applied, the case-bearing nominal is usually assumed to become inactive and, consequently, unable to receive additional case features later in the derivation.

At the same time, a number of recent proposals, often developed independently and for different empirical domains, have argued for alternative conceptions of case assignment. These approaches depart from the standard model in allowing a case feature to be assigned more than once, or to be otherwise modified after the initial case-assigning operation (Béjar and Massam 1999, Matushansky 2010, Pesetsky 2013, Richards 2013, Levin 2017, Ganenkov 2022 among others). The present paper contributes to this second line of research. We propose that case features initially assigned by Agree may be updated later in the derivation and that the calculation of the new case value is based on the decomposition of the case features and the superset relation between different case values.

The empirical basis for this proposal is Across-the-board (ATB) movement and the case-matching requirement associated with it in Russian. We show that Russian is, in general, well behaved with respect to case matching: in the overwhelming majority of contexts, the same case must be assigned in both conjuncts, or, if different cases are assigned, the conflict must be resolved by a syncretic form on the filler.

We show that the relevant exceptions arise in contexts involving the genitive of negation. Simplifying somewhat, the Russian genitive of negation may optionally replace accusative on direct objects, and it may also replace nominative on subjects of unaccusative verbs. A major new empirical contribution of this paper is the observation that a case mismatch in ATB-movement is grammatical when the genitive of negation replacing accusative in one conjunct is coordinated with accusative in the other conjunct. By contrast, the corresponding mismatch is ungrammatical when the genitive of negation replacing nominative in one conjunct is coordinated with nominative in the other conjunct.

We argue that this empirical pattern can be derived only under an asymmetric derivational approach to ATB-movement, in which the filler begins its derivational life in the second conjunct and then moves into the first conjunct (cf. the sideward movement account by Nunes 2004). This approach is combined with the idea that, when the nominal passes through a second case position, it can interact with the case feature available there, even though it already bears a valued case feature. The asymmetry between nominative- and accusative-based genitive of negation is derived by adopting an approach in which traditional case labels are not theoretical primitives, but are instead decomposed into more basic features. We suggest that genitive is built directly on the features of accusative, but does not stand in a full superset relation to nominative; cf. Jakobson (1936/1984), Neidle (1988), Halle (1997), Akkuş et al. (2025) for approaches to case decomposition involving only overlapping feature specifications as well as Calabrese (2008), Harðarson (2016), Christopoulos and Zompì (2022) for proposals involving feature decomposition with partial but not full case containment.

The paper is organized as follows. Section 2 provides the background necessary for the discussion, introducing ATB-movement, theories of ATB-movement, and the genitive of negation in Russian. Section 3 examines ATB-movement in Russian and presents new data on non-syncretic case mismatches. Section 4 develops the analysis of case assignment and case decomposition, and shows how, together with the assumptions about the syntax of ATB-constructions, it derives the Russian data. Section 5 considers additional contexts involving i.a. the lexical genitive and shows how they support the proposed analysis. Sections 6 and 7 explore the consequences of the proposal for theories of the Russian genitive of negation and for approaches to ATB-movement. Section 8 concludes.

2 Background

In this introductory section, we will provide the necessary background on the two phenomena that constitute the empirical core of the paper. We will first introduce ATB-movement and the constraints associated with it, and then the genitive of negation.

2.1 ATB-movement and the Case-matching requirement

2.1.1 The Coordinate Structure Constraint and ATB-movement

It has been known since Ross (1967) that extraction interacts with coordination in peculiar ways. On the one hand, extraction from a single conjunct of a coordination is ungrammatical, irrespective of conjunct order, see (1):

(1) a.*What did [John buy __] and [Mary sell magazines]?
b.*What did [John buy magazines] and [Mary sell __]?

To address this restriction, the Coordinate Structure Constraint (CSC) was proposed in Ross (1967), which essentially restates the observations illustrated in (1). On the other hand, extraction from a coordination *is* possible if extraction targets all conjuncts simultaneously, i.e., if there is a gap in each conjunct. Extraction of this type is referred to as *Across-the-board* movement and is shown in (2):

(2) What did [John buy __] and [Mary sell __]?

How exactly the CSC should be conceived of in modern times and why ATB-movement can circumvent it has been a controversial topic. It seems clear by now that the CSC is not exclusively a constraint on extraction; rather, in pretheoretical terms, it seems to be a more general requirement for the filler to establish links with each conjunct, but those links need not necessarily arise via movement, they can also do so via binding, see Ruys (1992), Fox (2000), Lin (2002), Salzmann (2012*b*) and the overview in Sportiche (2024). For a recent approach that tries to subsume the CSC under Phase Theory, see Bošković (2020).

Irrespective of the precise implementation of the CSC, ATB-movement presents a challenge to any syntactic approach in that there seems to be a mismatch between the number of fillers (1) and the number of traces (2). It seems as if there are two instances of extraction that somehow result in just a single filler.[1] Various solutions to this challenge have been proposed.

For the purposes of this paper, they can be classified along two lines: First, how the filler is related to the conjuncts, and second, which is largely a consequence of the first, what degree of similarity is predicted for the representations of the filler inside the conjuncts. Regarding the first parameter: in some approaches, the filler is directly, and simultaneously, related to both gaps; this is the case in certain forms of argument sharing in earlier approaches, (e.g., Goodall 1987, Moltmann 1992) and multidominance accounts in more recent work (e.g., Citko 2005, Bachrach and Kazir 2009, Gračanin-Yuksek 2013, Citko and Gračanin-Yuksek 2021, Hewett 2025), as well as in parallel extraction accounts (Williams 1978, Blümel 2017, Hein and Murphy 2020).[2] In other approaches, the relationship between fillers and gaps is indirect in that there is only extraction from one conjunct, while the gap in the other conjunct comes about in a different way. In the empty operator approach (Munn 1993, Franks 1995, Bošković and Franks 2000), which treats parasitic gaps and ATB-movement on a par, the gap in the non-initial conjunct arises from empty operator movement, while the filler is extracted from the first conjunct. In ellipsis-approaches, there is a representation of the filler in both conjuncts, but it undergoes ellipsis in one of the conjuncts, under identity with the filler in the other conjunct. There is thus extraction from only one conjunct, either the first (forward ellipsis, Salzmann 2012*a*) or the second (backward ellipsis, Ha 2008).[3] The last type of approach to be discussed is the sideward movement approach (Nunes 2004). In this approach, the filler is initially merged in the second conjunct. From there it is copied to the first conjunct before the two conjuncts are merged (hence an instance of sideward movement). Subsequently, the filler is extracted from the first conjunct. The approach therefore also qualifies as asymmetric.

Turning to the second parameter, since semantically, the filler is related to both conjuncts,

[1]We take it for granted that ATB-examples like the ones above cannot be reanalyzed as instances of full CP-coordination plus ellipsis of the filler (and potentially more material) in the second conjunct. The relevant parse can be ruled out in various ways and specific constraints on ATB-movement such as case-matching and parallelism (see below) additionally argue against such a possibility.

[2]Non-derivational approaches like Pollard and Sag (1994) also belong in this category: they involve simultaneous slash-feature percolation from both conjuncts, which are merged into a single slash feature via set union at the &P-level.

[3]Two further approaches based on asymmetric extraction are Zhang (2010), where there is an empty pro in the second conjunct in the position the filler relates to (and no movement), and Larson (2013), where there is no representation of the filler at all in the second conjunct. We will not discuss these approaches any further because they do not seem to offer straightforward possibilities to implement case-matching and in addition fail to account for movement effects inside the second conjunct, which are, however, generally attested when the second conjunct contains a gap.

there needs to be a representation of the filler in each conjunct. But the approaches differ in how similar they predict those representations to be. Symmetrical approaches essentially predict identity: In sharing/multidominance approaches, there is only one representation of the filler inside both conjuncts, hence the representations in two conjuncts are expected to be identical (modulo differences due to syncretism, see below). The same holds for the parallel extraction account where the fusion of the two filers depends on identity of features (again, modulo differences due to syncretism). Things are different in the empty operator movement account. To ensure that the filler is related to the second conjunct, the operator movement chain needs to be linked to the Ā-chain in the matrix clause, which usually arises via some form of chain composition. To the extent that this mechanism has been spelled out, it mainly seems to predict identity regarding semantic properties but not necessarily morphosyntactic features. In ellipsis approaches, the extracted operator binds the variable in the second conjunct that results from movement of the other instance of the filler in that conjunct. This ensures that filler is interpreted in both conjuncts. But since identity is regulated by ellipsis, which is known to tolerate mismatches (especially under a semantic identity condition), the approach does not predict identity in morphosyntactic features; in fact, it also allows for more substantial mismatches such as those resulting from vehicle change. Finally, in the sideward movement approach, which involves copying from one conjunct to the other and chain formation of the filler with all copies inside the conjuncts, we essentially expect the representations of the filler to be identical, though depending on the implementation, certain mismatches are possible (see below).

Against the background of these different implementations of ATB-movement, we will now turn to established constraints on ATB-movement.

2.1.2 Case-matching in ATB-movement

ATB-movement itself is also subject to specific constraints. The most important one is that, cross-linguistically, ATB-movement is subject to a case-matching requirement, as first discussed in Dyła (1984) for Polish. The requirement is illustrated in (3-a-b) with Polish data. In the first example, one and the same case, namely accusative, is assigned to the gap position in both conjuncts. The wh-pronoun likewise realizes accusative, and the sentence is grammatical. The example in (3-b) forms a minimal pair with (3-a) and differs from it only in that dative is assigned in the second conjunct. Such ATB-movement is ungrammatical regardless of whether the wh-pronoun shows accusative or dative case. Here and throughout the paper, cases assigned in the first and in the second conjunct are indicated after the English translation of an example.

(3) a. Kogo Jan lubi __ a Maria podziwia __?
who.ACC Jan likes and Maria admires
'Who does Jan like and Maria admire?' ACC & ACC

b. *Kogo / *Komu Jan lubi __ a Maria ufa __?
who.ACC / who.DAT Jan likes and Maria trusts
'Who does Jan like and Maria trust?' Citko (2005: 485) ACC & DAT

The only known way to resolve a conflict between two non-matching cases is by means of case syncretism; see, e.g., Borsley (1983), Dyła (1984), Franks (1993, 1995), Bondaruk (2003), Citko (2005, 2011), Hartmann et al. (2016), Hein and Murphy (2020), Citko and Gračanin-Yuksek (2021). For Polish this is illustrated in (4-a-b). Both examples show a mismatch between the accusative assigned in the first conjunct and the genitive in the second conjunct. Nevertheless, example (4-a) is grammatical because the form of the wh-pronoun is syncretic between accusative and genitive, and informally speaking can fulfill the case requirements of both conjuncts. The form of the inanimate pronoun in (4-b) is not syncretic: *co* realizes only accusative, the genitive form being *czego*.

As a result, it cannot resolve the mismatch between accusative and genitive, and the sentence is ungrammatical.

(4) a. Kogo Janek lubi __ a Jerzy nienawidzi __?
who.ACC/GEN John likes and George hates
'Who does John like and George hate?' ACC & GEN

b.*Co Janek lubi __ a Jerzy nienawidzi __?
what.ACC John likes and George hates
'What does John like and George hate?' Dyła (1984: 701-702) ACC & GEN

In addition to case-matching, ATB-movement requires some sort of parallelism. Its exact nature is still very much debated. Sometimes, it is taken to be structural parallelism, i.e., the extraction sites must be in the same structural position, see, e.g., Hein and Murphy (2020), Citko and Gračanin-Yuksek (2021). This has been proposed to explain why examples like (5) are ungrammatical despite the filler being syncretic (Williams 1978: 34):

(5) *I know a man who [Bill saw __] and [__ likes Mary].

If extraction takes place from Spec,vP in the first and Spec,TP in the second conjunct, the parallelism requirement is violated in (5), which is therefore correctly predicted to be ungrammatical.

All the examples in this paper satisfy the structural parallelism requirement: All extractions take place from underlying object position in the broad sense, including the base position of datives and oblique objects. Even if those positions may be somewhat different in that, e.g., some objects are extracted from the complement position of V, while others (depending on one's analysis) may be extracted from a higher position such as Spec,ApplP, as, e.g., in the gen-dat syncretism in (16) below, the extraction sites will count as identical if there is first intermediate movement via Spec,vP before Ā-movement displaces the filler to Spec,CP.

In other work, what is taken to be at stake is parallelism regarding thematic prominence (Franks 1993: 515) such that

(6) "In any ATB construction, the gaps must pertain either to most prominent or to not most prominent arguments, consistently across all the conjuncts."

This was proposed to account for examples that were ungrammatical even though they met the (morphological) case-matching requirement.[4] In these examples, one of the gaps corresponds to the sole argument and hence the most-prominent argument of a verb, while the other gap corresponds to an object that is less prominent than the other argument of the verb (usually agent or experiencer). At least some of the examples discussed in Franks (1993) potentially constitute a challenge for structural parallelism in that they arguably satisfy parallelism but are nevertheless ungrammatical. For instance, dative experiencers and dative objects are often assumed to be projected in the same position (SpecApplP), hence combining them in ATB-movement should not be a problem and the presence of a higher argument above the dative objects should not affect parallelism. In probably all (the other) examples, the arguments are internal arguments.[5] Even if the

[4]This involves (i) examples combining dative objects with dative experiencers (see, e.g., Dyła 1984: 704, ex. 17, Franks 1993: 513, ex. 9/11), (ii) examples that (syncretically) combine a dative object and a genitive of negation on an unaccusative subject (Dyła 1984: 704, ex. 16, Franks 1993: 513, ex. 10), (iii) examples that (syncretically) combine an accusative experiencer with a genitive object (Franks 1993: 518, ex. 17b/20b), (iv) examples that syncretically combine a nominative intransitive (probably unaccusative) subject with an accusative object (Franks 1993: 518:21b), and (v) examples that (syncretically) combine a genitive unaccusative subject with an accusative theme of an experiencer verb (Franks 1993: 519, ex. 22c).

[5]In the case of (iv) of the previous footnote, both are arguably complements of V and hence in parallel position, the same may hold for (iii) and (v).

base-position may not be fully identical as in cases where extraction of an indirect object is combined with extraction from the complement of V position, intermediate movement via Spec,vP would void any structural differences (recall the discussion regarding ex. (16) above).[6]

There are reasons to believe, though, that the thematic approach cannot account for the entire range of possible (mis-)matches; see, e.g., Bondaruk 2003: 236, ex. 30 for an example that satisfies Franks' prominence condition (dative experiencer combined with genitive unaccusative subject) but is ungrammatical nevertheless. Some of the data discussed in our paper are also not compatible with Franks' approach. For example, the examples in (14) below combine a direct object with an unaccusative subject and hence a not most prominent argument with a most prominent argument, but they are nevertheless grammatical. The same holds for examples (38), (41), and (44) below. Thus, while all are compatible with structural parallelism, they are not compatible with thematic prominence. We will not attempt to resolve this issue here as we believe that it is orthogonal to the goals pursued in this paper.

Rather, the focus in what follows will be to show that Russian allows non-syncretic mismatches in one very specific case, namely when the genitive of negation is involved.

2.2 The genitive of negation

The second phenomenon that is at the center of this paper is the genitive of negation. In Russian as in some other (Balto-)Slavic languages, the regular case that normally appears on an argument is replaced by genitive in the presence of sentential negation. This genitive is referred to as the genitive of negation. Russian differs from languages like Polish or Lithuanian in that genitive of negation is optional. Descriptively, genitive of negation only occurs on internal arguments that are assigned structural case; see Harves (2013) for an overview of the discussion and section 6 below for empirical refinements. Depending on the case that is replaced by genitive of negation, two cases are typically identified.

First, genitive of negation appears on direct objects of transitive verbs, replacing accusative case. This is shown in (7): Example (7-a) demonstrates that genitive is ungrammatical on the regular direct object in the absence of negation; it becomes possible in (7-b) where the sentential negation is added.

(7) a. Anna kupila žurnal / *žurnala.
Anna bought magazine.ACC magazine.GEN
'Anna didn't buy a magazine.'
b. Anna ne kupila žurnal / žurnala.
Anna NEG bought magazine.ACC magazine.GEN
'Anna didn't buy a magazine.' Harves (2013: 647)

Second, genitive of negation can replace nominative where two subcases can be distinguished: subjects of lexical verbs and subjects of copular constructions. Starting with lexical verbs, genitive of negation is restricted to subjects of unaccusative and passive verbs and does not affect unergative or transitive subjects. The data in (8-b) illustrate genitive of negation on an unaccusative subject, while (8-a) confirms that genitive is ungrammatical on this type of subject if negation is not present. Note that the two examples also differ regarding verb agreement. The verb agrees with the nominative subject and shows masculine gender inflection in (8-a), but agreement turns out to be impossible for the genitive subject in (8-b) and the verb surfaces with the default neuter inflection

[6]The configurations discussed in fn. 4 can arguably only be ruled out on the basis of structural parallelism if the sole argument necessarily moves to Spec,TP and is extracted from there. While this may be plausible with the experiencers in (i) and (iii), this seems unlikely for the unaccusative subjects in (iv) and (v).

instead (agreement in Russian is case-discriminating and restricted to nominative).[7]

(8) a. Otvet / *otveta iz polka ne prišël.
answer.NOM answer.GEN from regiment NEG arrived.M.SG
'The answer from the regiment did not arrive.'
b. Otveta iz polka ne prišlo.
answer.GEN from regiment NEG arrived.N.SG
'No answer from the regiment arrived.' Harves (2013: 647)

Turning now to copular constructions, in the presence of negation, genitive may replace nominative on the subject of the verb *byt'* 'to be' in existential, locative, and possessive sentences. The alternation is illustrated in (9) with an existential clause: the data in (9-a) show that a genitive subject is ungrammatical in the corresponding affirmative clause, whereas (9-b) shows that it becomes grammatical under negation. Analogous to (8-b), the verb here cannot agree with the genitive subject and shows default neuter agreement.

(9) a. Na uroke byl Pa'vsa / *Paši.
at lesson be.M.SG Pasha.NOM Pasha.GEN
'Pasha was in class.'
b. Na uroke ne bylo Paši.
at lesson NEG be.N.SG Pasha.GEN
'Pasha was not in class.'

Unlike in some other Slavic languages such as Lithuanian (Sigurðsson and Šereikaitė 2024) and Polish (Przepiórkowski 2000, Witkoś 2008), there are semantic factors associated with the genitive of negation in Russian (see Babby 1980, Partee and Borschev 2004); i.e., it is not the case that any NP that fulfills the criteria above will necessarily end up bearing genitive. The exact conditions for the appearance are extremely hard to pin down, differ between accusative and nominative-based genitive of negation (see Harves 2013) and also seem to be subject to a high degree of inter-speaker variation. Simplifying somewhat, the genitive of negation tends to be associated with non-specific indefinite interpretation / to come without an existence presupposition (but it is by no means restricted to those contexts). These semantic factors will not a play a role in what follows and our analysis will not attempt to implement these restrictions. Instead, the main data discussed here aim to keep the relevant semantic parameters constant, thereby ensuring that they cannot be responsible for the observed pattern.

3 Data

We begin this section by introducing ATB-movement in Russian and showing that the language is generally well behaved with respect to the case-matching condition. We then turn to the genitive of negation and demonstrate that it gives rise to grammatical non-syncretic mismatches.

3.1 ATB-movement in Russian

Unlike Polish, for which ATB-movement is comparatively well documented and has played an important role as a testing ground for theories of ATB-dependencies, Russian ATB-movement has received substantially less attention (see, however, Franks 1993, 1995 for the basic Russian data and for comparison with Polish). The example in (10) illustrates a case of ATB-movement in Russian

[7] Verbs in Russian show person agreement in the present tense, but only gender and number agreement in the past tense that is used in these examples.

wh-questions. In this example, the wh-NP *kakogo aktera* 'which actor' is linked to direct object gaps in both conjuncts. Accusative case is assigned to the gap position in each conjunct, and the wh-phrase likewise shows accusative case marking, thereby satisfying the case-matching requirement.[8]

(10) [Kakogo aktëra] Ira ljubut __ i Galja obožaet __?
which.ACC actor.ACC Ira loves and Galya adores
'Which actor does Ira love and Galya adore?' ACC & ACC

The examples below provide further grammatical instances of ATB-movement involving full case matching. In (11-a), dative is the case realized by the filler and assigned in both gap positions, and in (11-b) it is instrumental.

(11) a. Kakomu studentu Maša pozvonila __ i Petja napisal __?
which.DAT student.DAT Masha called and Petya wrote
'Which student did Masha call and Petya write to?' DAT & DAT
b. Kakim proektom Ivan rukovodit __ i Maša zanimaetsja __?
which.INS project.INS Ivan supervises and Masha works.on
'Which project does Ivan supervise and Masha work on?' INS & INS

The examples in (12)-(13) illustrate violations of the case-matching requirement and are therefore ungrammatical. The examples in (12) involve a mismatch between instrumental and accusative, whereas those in (13) involve a mismatch between dative and accusative. The [a] and [b] variants differ only in the order of the conjuncts, thereby showing that the mismatch is ruled out independently of which case is assigned in which conjunct.

(12) a.*[Kakoj mašinoj] Ira pol'sovalas' __, a Galja kupila __?
which.INS car.INS Ira used but Galya bought
'Which car did Ira use but Galya buy?' INS & ACC
b.*[Kakuju mašiny] Galja kupila __ i Ira pol'sovalas' __?
which.ACC car.ACC Galya bought and Ira used
'Which car did Galya buy and Ira use?' ACC & INS

(13) a.*[Kakomu studentu] Maša zvonit __ i Petja iš'č'et __?
which.DAT student.DAT Masha calls and Petja looks.for
'Which student is Masha calling and Petja looking for?' DAT & ACC
b.*[Kakogo studenta] Petja iščet __ i Maša zvonit __?
which.ACC student.ACC Petja looks.for and Masha calls
'Which student is Petja looking for and Masha calling?' ACC & DAT

Finally, similarly to other languages, case mismatches in Russian can be resolved by syncretism, i.e., the assignment of distinct cases in the two conjuncts can be tolerated, but only when the wh-phrase surfaces in a form syncretic between the two conflicting cases.

Depending on the part of speech (noun vs. adjective), gender, number, and declension class, the following case syncretisms are attested in Russian: Nom-Acc, Acc-Gen, Dat-Loc, Gen-Dat-Loc.[9] The data in (14)-(16) show that syncretic Nom-Acc, Acc-Gen, Gen-Dat mismatches are grammatical. Syncretisms involving Loc are not relevant for our purposes, because the case is only assigned by prepositions and Russian has no preposition stranding, so that ATB-movement

[8]Critical data points without reference were constructed by the authors and were judged by at least 5 Russian native speakers.

[9]Gen-Dat-Loc-Ins syncretism is attested in the adjectival declension, including the relative pronouns (see, e.g., Franks 1993: 513).

of locative NPs is ungrammatical for independent reasons.

(14) a. Kakoe pis'mo ležalo __ na stole i Ivan pročital __?
which.N.NOM/ACC letter.NOM/ACC laid on table and Ivan read
'Which letter was lying on the table and did Ivan read?' NOM & ACC

b. Kakoe pis'mo Ivan pročital __ utrom i večerom
which.N.NOM/ACC letter.NOM/ACC Ivan read in.the.morning and in.the.evening
eščo ležalo __ na stole?
still laid on table
'Which letter did Ivan read in the morning and was still lying on the table in the evening?'
ACC & NOM

(15) a. [Kakogo druga] Petja lišilsja __, a Maša priobrela __?
which.M.ACC/GEN friend.ACC/GEN Petja lost and Masha gained
'Which friends did Petja lose and Masha gain?' GEN & ACC

b. [Kakogo druga] Maša priobrela __, a Petja lišilsja __?
which.M.ACC/GEN friend.ACC/GEN Masha gained and Petja lost
'Which friend did Masha gain and Petja lose?' ACC & GEN

(16) a. [Kakoj lošadi] Maša eščë ne videla __, a Ira uže dala __ edu?
which.F.GEN/DAT horse.GEN/DAT Masha yet not saw but Ira already gave food
'Which horse has Masha not seen yet and Katja already given food to?' GEN & DAT

b. [Kakoj lošadi] Ira uže dala __ edu, a Maša eščë ne videla __?
which.F.GEN/DAT horse.GEN/DAT Ira already gave food, but Masha yet not saw
'Which horse has Ira already given food to and Masha not seen yet?' DAT & GEN

The discussion so far shows that Russian initially patterns in a fully expected way with respect to the case-matching requirement. ATB-movement is grammatical when the same case is assigned in the two conjuncts, ungrammatical when distinct non-syncretic cases are assigned, and grammatical again when the mismatch is resolved by syncretism on the filler.

3.2 Non-syncretic mismatches with the genitive of negation

We now introduce genitive of negation into the picture and provide new data from Russian showing that ATB-movement is grammatical with non-syncretic mismatches if the genitive of negation is involved, but crucially only in certain configurations, i.e., in certain conjunct orders.

In the first configuration, the filler is a direct object in both conjuncts, with both verbs assigning accusative case. Here, a non-syncretic mismatch is possible with a genitive filler if sentential negation is located in the first conjunct, (17-a), but not if it occurs in the second, (17-b):[10]

(17) a. [Kakix knig] Katja daže ne videla __, a Maša uže čitala __?
which.PL.GEN book.PL.GEN Katja even not saw but Masha already read
'Which books has Katja not even seen but Masha already read?' GENNEG(<ACC) & ACC

b.*[Kakix knig] Maša uže čitala __, a Katja daže ne videla __?
which.PL.GEN book.PL.GEN Masha alrealy read but Katja even not saw
'Which books has Masha already read but Katja not even seen?' ACC & GENNEG(<ACC)

[10] Non-syncretic mismatches involving the genitive of negation have been previously documented for Polish, see Bondaruk (2003: 231f.), Rothert (2022). Both find that that mismatches are grammatical in the GenNeg(<Acc)–Acc configuration as long as the filler matches the case requirements of the first conjunct, while a filler matching the requirements of the second conjunct is judged as ungrammatical/degraded. Bondaruk (2003: 231f.) additionally claims that the Acc-GenNeg(<Acc) order is grammatical with an accusative filler, but not with a genitive filler. Neither of the papers provides an analysis of the facts; in addition, ATB-examples with the genitive of negation based on nominative (as in (19)) and the additional contexts we discuss in section 5 below are not addressed.

Since the genitive of negation in Russian, unlike in some other Slavic languages, is never obligatory, the sentences in (18), which differ from those in (17) only in that the filler bears accusative case, are grammatical. This confirms that the ungrammaticality of (17-b) should indeed be traced to the case mismatch between genitive on the filler and accusative assigned in the first conjunct. At the same time, the accusative data in (18) are uninformative for investigating which other case mismatches may be grammatical, since there is no way to ensure that the genitive of negation has in fact been assigned in the negated conjunct.[11]

(18) a. [Kakie knigi] Katja daže ne videla __, a Maša uže čitala __?
which.PL.ACC book.PL.ACC Katja even not saw but Masha already read
'Which books has Katja not even seen but Masha already read?' ACC & ACC

b. [Kakie knigi] Maša uže čitala __, a Katja daže ne videla __?
which.PL.ACC book.PL.ACC Masha alrealy read but Katja even not saw
'Which books has Masha already read but Katja not even seen?' ACC & ACC

So far it seems that non-syncretic mismatches are grammatical if the filler meets the requirements of the first conjunct, where one expects the direct object to surface with genitive. Note that the effect in (17) cannot be reduced to a general proximity effect that is often observed in coordination (regarding phenomena such as agreement or binding) because non-syncretic mismatches involving other cases are ungrammatical, recall examples (12) and (13) from above.

Against this background, one may expect similar mismatches to be possible with the genitive of negation on subjects. Surprisingly, however, a fact that has not been previously observed in any Slavic language, this expectation is not borne out. Non-syncretic mismatches where a genitive on an unaccusative subject is combined with nominative on an unaccusative subject is impossible, irrespective of the order:

(19) a.*Kakix pisem eščë ne prišlo __, no byli otpravleny __ nedelju nazad?
which.PL.GEN letter.PL.GEN yet not arrived but were sent week ago
'Which letters have not arrived yet, but have been sent a week ago?'
GENNEG(<NOM) & NOM

b.*Kakix pisem byli otpravleny __ nedeju nazad, no eščë ne prišlo __?
which.PL.GEN letter.PL.GEN were sent week ago not yet not arrived
'Which letters have been sent a week ago, but have not arrived yet?'
NOM & GENNEG(<NOM)

The examples in (20) constitute a minimal pair with the data in (19) and differ only in that the filler realizes nominative case. These examples confirm that the ungrammaticality of both examples in (19) is due to the mismatch between nominative and genitive.

(20) a. Kakie pis'ma eščë ne prišli __, no byli otpravleny __ nedelju nazad?
which.PL.NOM letter.PL.NOM yet not arrived but were sent week ago
'Which letters have not arrived yet, but have been sent a week ago?' NOM & NOM

b. Kakie pis'ma byli otpravleny __ nedeju nazad, no eščë ne prišli __?
which.PL.NOM letter.PL.NOM were sent week ago not yet not arrived
'Which letters have been sent a week ago, but have not arrived yet?' NOM & NOM

The final context where genitive of negation is grammatical are subjects of the verb *byt'* 'to be', and they show the same pattern as unaccusative subjects of lexical verbs: a genitive filler is ungram-

[11] One possibility to test accusative fillers is by using lexical genitives, which are not optional. Those are ungrammatical with an accusative filler when combined with accusative case in the other conjunct, irrespective of the conjunct order, see section 5.4.

matical when genitive on the subject is assigned in only one conjunct, while the other conjunct requires nominative.

(21) a. *Kakogo studenta __ ne bylo na uroke i isčez __ na prošloj nedele?
which.GEN student.GEN NEG was at lesson and disappeared on last week
'Which student was not in class and disappeared last week?' GENNEG(≺NOM) & NOM
b. *Kakogo studenta isčez __ na prošloj nedele i ne bylo __ na uroke?
which.GEN student.GEN disappeared on last week and NEG was at lesson
'Which student disappeared last week and was not in class?' NOM & GENNEG(≺NOM)

The data in (22) provide minimal pairs to these examples, confirming that the ungrammaticality arises from the case mismatch.

(22) a. Kakoj student __ ne byl na uroke i isčez __ na prošloj nedele?
which.NOM student.NOM NEG was at lesson and disappeared on last week
'Which student was not in class and disappeared last week?' NOM & NOM
b. Kakoj student isčez __ na prošloj nedele i ne byl __ na uroke?
which.NOM student.NOM disappeared on last week and NEG was at lesson
'Which student disappeared last week and was not in class?' NOM & NOM

Thus, at a pretheoretical level, the genitive of negation on a direct object seems to count as an accusative regarding the case matching requirement, while the genitive of negation on a nominative—unexpectedly—does not count as a nominative. This key difference will motivate crucial aspects of our analysis which is to be introduced in the next section.

4 Analysis

This section develops an analysis of the apparent asymmetry between accusative-based and nominative-based genitive of negation. We first present our proposal for case decomposition with partial case containment, as well as the mechanism of multiple case assignment in 4.1 and then turn to the analysis of ATB-movement in 4.2. After this, we show how our analysis captures the case matching requirement in 4.3 and finally present explicit derivations of grammatical non-syncretic mismatches involving genitive of negation in 4.4.

4.1 Case decomposition and multiple case assignment

The central theoretical innovation of this paper lies in the domain of case and consists, more specifically, of two ingredients: a mechanism of multiple case assignment and a theory of case decomposition. In this subsection, we introduce both of these ingredients.

We begin by spelling out the mechanism of case assignment. In line with standard minimalist approaches to case assignment, we assume that case assignment proceeds via Agree. For concreteness' sake, we take the unvalued case feature on the noun to function as the probe, while the clausal heads traditionally referred to as case assigners function as the corresponding goals.[12]

[12]Strictly speaking, Chomsky's original view is that case assignment is a by-product of ϕ-agreement (Chomsky 2000, 2001). Although it has since been shown that case assignment need not be tied to ϕ-agreement at least on the surface, this broad idea of case as a by-product can also be interpreted in a different way: A feasible alternative is a system in which clausal heads bear valued case probes, while nominals bear unvalued case goals.

Such a reversal of probe and goal relations would have relatively little effect on the actual mechanics of the present analysis. It would remain fully compatible both with the feature decomposition proposed in this subsection and with the syntax of ATB-movement outlined in the following subsection. The challenge for the alternative described in this footnote, as for the one adopted in the main text, is to make it possible for a single nominal to receive multiple case

On its standard formulation, however, Agree is commonly defined as a relation between a single probe and a single goal, whereby the unvalued feature of the probe is valued. Once this relation has been established, the probe is typically assumed to become inactive and, as a result, is no longer able to enter into further Agree relations with additional goals.

We depart from this assumption and suggest that after a nominal has received a case value, it can still interact with new case probes. This proposal is not without precedent. Rather, it is in line with a growing body of work that explores the possibility that a single nominal may receive more than one case value in the course of the derivation (see Béjar and Massam 1999, Bianchi 2000, Matushansky 2010, Spyropoulos 2011, Pesetsky 2013, Richards 2013, Deal 2016, Levin 2017, Fong 2019, Ganenkov 2022 among others).[13]

We propose that a nominal is required to enter into an agreement relation with a new case assigner whenever it occurs in a position where that case is normally assigned. We suggest that this can be technically implemented by minimally modifying the standard definition of Agree; specifically, by removing the requirement that a probe must always bear an unvalued feature in order to interact with a new goal (cf. Béjar and Rezac 2009, Deal 2015, Coon and Keine 2021 for other approaches allowing a probe to interact with multiple goals[14]).

As a result, if a nominal occupies several case-assigning positions in the course of the derivation, it is expected to accumulate case features from all of them. Case assignment then has a broader domain of application, and it must be constrained by restrictions on a single goal participating in multiple Agree relations.[15]

values. As in the system here, this can be achieved by assuming that, after receiving case, a nominal may acquire additional case features if it appears in another case-assigning position.

[13]One commonly adopted condition that renders a nominal inactive after case assignment is the Activity Condition (Chomsky 2000). It requires a nominal to bear an unvalued case feature in order to be able to participate in A-related Agree operations (viz., agreement or case assignment). However, a substantial body of work has independently shown that the Activity Condition cannot be maintained as a cross-linguistically valid constraint. This is particularly clear in cases where an NP interacts with several Agree probes as, e.g., in long-distance agreement where an NP receives case and agrees with a probe in an embedded clause but is nevertheless eligible to agree with another probe in the matrix clause (see, Bhatt and Keine 2017 for an overview and references). The same issue can arise within the same clause when an NP is first assigned a case lower in the structure (e.g., ergative) before agreeing with the ϕ-probe on T. Another phenomena potentially at odds with the Activity Condition is hyperraising where a subject raises from a finite clause and triggers agreement with both the embedded and the matrix T (and, depending on one's analysis, may also receive two cases), see, e.g., Halpert (2019). On the other hand, there are languages where the Activity Condition seems to hold. As a consequence, it is sometimes treated as a parameter (see, e.g., Baker 2008).

[14]Both Béjar and Rezac (2009) and Deal (2015) posit additional conditions on when and how a probe may interact with a second goal. For present purposes, however, it is important not to impose any such condition: the second case feature is acquired and may affect the outcome of the derivation independently of the relation between the two case values.

[15]This modification raises the question of whether such a system can still assign all cases within a clause correctly, or whether it makes incorrect empirical predictions by allowing a nominal to acquire more case features than it normally would in an ordinary clause. In this footnote, we take on the task of addressing this question and show that the proposed modifications still derive the correct pattern of case assignment within the clausal domain.

Let's consider a simple ditransitive clause and assume that it has the structure $[_{TP}$ T $[_{vP}$ SU v $[_{ApplP}$ IO Appl $[_{VP}$ V DO $]$ $]$ $]$ $]$. This structure contains three unvalued case probes, located on the three nominals—the indirect object, the direct object, and the subject—and three valued case goals: dative on Appl, accusative on v, and nominative on T.

Case assignment proceeds as follows. First, once ApplP has been built, the structure contains two unvalued case probes—one on the direct object and one on the indirect object—and one case goal, located on Appl. We assume that Spec-Head Agree takes priority in this case, so that the indirect object receives dative case. This assumption, however, is not specific to the present analysis: in one way or another, any theory of case assignment must ensure that dative is assigned to the indirect object rather than to the direct object.

Next, v is merged. Since, on the present approach, a nominal probe may receive additional case values even after an initial case value has been assigned, both the indirect object and the direct object are in principle eligible to receive accusative at this point. We assume that both probes can locate the accusative goal, but that only one of them can receive the case feature, since a case goal may enter into only one case relation. If the direct object receives the case

Table 1: Feature specifications of Russian cases

case	*feature*		*syncretisms*	
NOM	α		NOM	[Ø]
ACC	β	$[\beta]$	ACC	[Ø]
GEN	β,γ	$[\beta]$	GEN	$[\gamma]$
DAT	γ,δ,ϵ	$[\gamma,\delta]$	DAT	$[\gamma]$
LOC (PREP)	γ,δ,ζ	$[\gamma,\delta]$	LOC	$[\gamma]$
INS	γ,η		INS	$[\gamma]$

The second ingredient of the proposal concerns the nature of the interaction between several case values assigned successively to the same nominal in the course of the derivation. This is where case decomposition and containment become relevant.

We assume that traditional case labels such as nominative, accusative, and genitive should not be treated as theoretical primitives. Rather, they are decomposed into combinations of more basic privative features. Crucially, we propose that there is partial case containment, so that the accusative is featurally contained in the genitive, while there is no containment among other cases, i.e., neither accusative nor genitive contains nominative. Concretely, we suggest the case specification for the Russian cases given in Table 1.

Our feature decomposition is motivated both syntactically, i.e., by the behavior in ATB-movement, and morphologically, i.e., syncretism. The lack of full containment between cases in our approach puts it in line with syntactically motivated case decompositions, under which cases typically have only overlapping feature specifications (see Neidle 1988, Akkuş et al. 2025). Our proposal therefore deviates from recent morphologically oriented work, most notably Caha (2009), where full containment is argued to be necessary in order to account for morphological syncretisms. It should be noted, however, that full case containment has never been, and still is not, necessary for an account of case syncretism; see Jakobson (1936/1984), Halle (1997), and Halle and Vaux (1998) for classical accounts without containment, as well as Calabrese (2008), Harðarson (2016), and Christopoulos and Zompì (2022) for more recent work arguing that weaker case containment is in fact better suited for capturing morphological syncretisms. Naturally, given the role of syncretism in alleviating case mismatches in ATB-movement, our decomposition is also compatible with these effects; see the derivations in section 4.3.

Having introduced the two core components of the analysis—multiple case marking and case decomposition—we now turn to how the two affect one another. We propose that once a case probe, i.e., an NP, has acquired case features from its first goal, i.e., from the clausal head, its subsequent interaction with another goal bearing a new case value is subject to an additional restriction: Case features can be added to a case feature set of an NP that was created in an earlier case assignment operation only if they are a superset of the earlier assigned features (cf. also Neidle 1988). If the features to be added only constitute a subset or are different from the existing feature set on an NP, a new feature set is created on the NP.

This means that if the noun phrase receives accusative $[\beta]$ and is subsequently assigned genitive $[\beta, \gamma]$, the new feature $[\gamma]$ is simply added to the case feature set already present on the noun:

feature, the derivation converges. If, however, it is the indirect object that receives the case value, the derivation crashes for two reasons. First, the direct object remains caseless and therefore violates the Case Filter. Second, the indirect object now bears too many case features, which creates a problem for its realization at PF; see the discussion later in this subsection as well as section 6.2.

Finally, the subject is introduced into the structure and for the derivation to converge it must receive case from the remaining case goal on T.

(23) a. [$X_{[\beta]}$ $NP_{[\beta]}$]
1

b. [$Y_{[\beta,\gamma]}$ $X_{[\beta]}$ $NP_{[\beta,\gamma]}$]
2 ✓

However, if the noun first receives genitive [β, γ] and is assigned accusative later in the derivation, a new feature set on the NP will be created because the later added feature [β] is not a superset of [β, γ]. In (24-b), and throughout the rest of the paper, we use the ✗ sign to indicate that the new features cannot be added to the existing feature set on the noun and that a new feature set is therefore created.

(24) a. [$X_{[\beta,\gamma]}$ $NP_{[\beta,\gamma]}$]
1

b. [$Y_{[\beta]}$ $X_{[\beta,\gamma]}$ $NP_{[\beta],[\beta,\gamma]}$]
2 ✗

Similarly, a new feature set will be created, for example, if genitive [β, γ] is assigned to an NP already bearing the nominative feature [α]:

(25) a. [$X_{[\alpha]}$ $NP_{[\alpha]}$]
1

b. [$Y_{[\beta,\gamma]}$ $X_{[\alpha]}$ $NP_{[\alpha],[\beta,\gamma]}$]
2 ✗

The distinction between one and two feature sets becomes crucial once the derivation reaches the morphological component. If a noun bears only a single set of case features, either because it receives only one case feature in the course of the derivation or because the second case is a superset of the first, this set is straightforwardly targeted by Vocabulary Insertion in the morphology. The best-matching exponent is then inserted in accordance with the Subset Principle (Halle 1997).

The situation is more complex when a noun bears two sets of case features at transfer to the morphology. Following Asarina (2011), Coon and Keine (2021), we assume that morphological realization then searches and identifies the best-matching exponent for each of the two feature sets. Since two exponents cannot be inserted into the same case slot, however, any attempt to do so causes the derivation to crash.

(26) Crash in morphology

$\text{noun}_{[\beta][\beta,\gamma]}$

VI: $[\beta,\gamma] \leftrightarrow$ A VI: $[\gamma,\delta,\epsilon] \leftrightarrow$ B

A noun bearing two sets of case features can be realized in the morphological component only if the best-matching exponent for each of the two feature sets is one and the same. In other words, the conflict between the two case values is resolved only when both are realized by a syncretic exponent.

(27) Resolution by syncretism

noun$_{[\beta][\beta,\gamma]}$

VI: $[\beta] \leftrightarrow$ A

Let us summarize the approach developed so far. We assume that case assignment proceeds via Agree, but depart from the assumption that the case probe (i.e., the NP) is deactivated once it has received a case value. This allows a noun to receive more than one case value if it occupies more than one case position in the course of the derivation. We further propose that a new case feature can be added to an existing feature set on the noun only if it is a superset of the previously assigned case feature. Otherwise, that is, if the newly assigned feature is not a superset of the existing one, a new feature set must be created on the noun.

At this point, this mechanism may seem arbitrary. We believe, however, that the rationale can be stated as follows: before new features can be added to an existing feature set, all features already present in that set must be matched against the newly assigned case features. Only then can the new features be introduced. This reasoning becomes more natural if features are represented as a hierarchical structure: new features can be added only at the top of the existing structure, and only if the resulting representation is well formed.

4.2 Implementation of ATB-movement

Concerning the implementation of ATB-movement, we adopt an asymmetric approach based on sideward movement (Nunes 2004), which can be sketched as follows: The basic idea is that the filler is merged and assigned case in the second conjunct; then, it moves to the first conjunct, where it receives another case. Finally, the filler undergoes Ā-movement to the final landing site in Spec,CP.

A few comments on the individual ingredients: The motivation for sideward movement is the familiar one, i.e., it comes from the fact that the predicate of the first conjunct is lacking an argument. This in turn requires there to be separate lexical arrays for each conjunct such that it is clear that one of the conjuncts is missing an argument. To satisfy the selectional requirements of the verb in the first conjunct, the filler undergoes sideward movement and merges with the V of the first conjunct. There, the filler will be in the purview of another case-probe and hence will be assigned another case; depending on the case feature, this will lead to the addition of a case feature to the existing feature set or a new feature set will be created (recall from above).[16] Finally, after the conjuncts have been merged with Conj and C′ is built, the filler undergoes Ā-movement from the first conjunct to the landing site outside of coordination. Depending on whether Vocabulary Insertion is possible into the position of the filler, the derivation either converges or crashes.

The treatment of ATB-movement in Nunes (2004) is somewhat brief and less explicit than that of parasitic gaps. We will therefore elaborate on certain aspects of the derivation that are important in what follows. First, as we will see in the derivations to come, it is crucial that sideward movement takes place from the second to the first conjunct. We propose that this can be done in a way similar to what is proposed in Nunes (2004: section 3.7) to ensure that sideward movement in parasitic gaps takes place from the adjunct clause to the argument position in the main clause and not vice versa (which would otherwise allow for ungrammatical instances of extraction from adjuncts): We

[16] We thus depart from the implementation in Nunes (2004) where case-checking takes place *after* Sideward Movement. He arguably makes this assumption to adhere to the Acitivity Condition, but as we are showing in this paper, multiple case assignment is instrumental in accounting for case matching. If there is no interaction between the cases assigned in the conjuncts, any kind of case mismatch is expected to be possible, not the least since one cannot require the copies to have identical case features; on this see fn. 18.

adopt a condition that a lexical array must be exhausted before a new array can be accessed. In addition, we assume that the coordinator is part of the array for the second conjunct (which it takes as its complement). This entails that that array must be complete: If it lacked an argument, the derivation would crash right away since it would have to be continued up to Conj′ before another array can be accessed. But if an argument is missing, the derivation will crash at the VP-/vP-level. Given that the array for the second conjunct is complete, sideward movement will only occur if the array for the first conjunct is missing an argument. If that is the case, sideward movement takes place from the second to the first conjunct. The following examples illustrate the possible arrays for a configuration where the direct object undergoes ATB-movement. We will assume that fillers consist of a quantifier and a noun, while the simple nominals are just N-heads. We will not address the question of whether noun phrases in Russian and Slavic more generally should be treated as NPs or DPs:

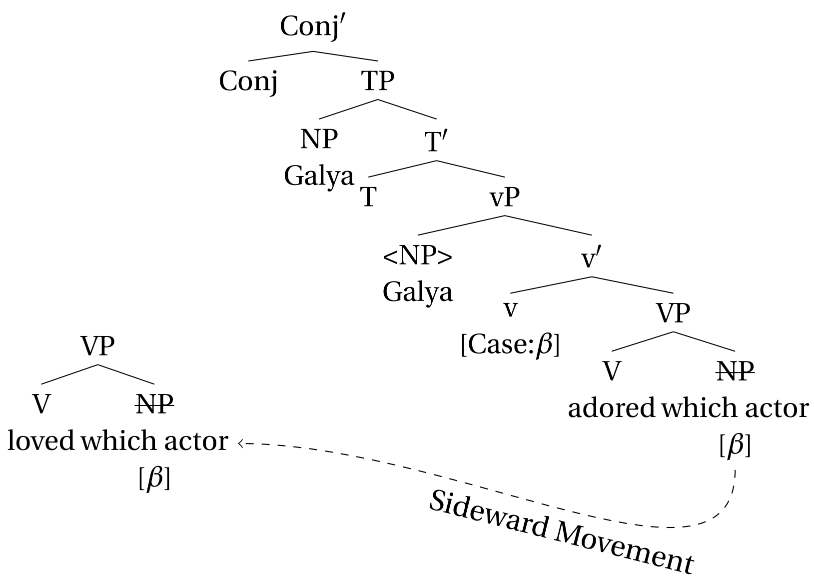
(28) arrays for conjoined TPs

a. {V, which, N, v, N, T, Conj} — second conjunct
b. {V, v, N, T} — first conjunct

Second, to ensure that the extracted filler realizes the combined features assigned by both verbs, extraction from coordination must take place from the first conjunct. This can be made to follow if the coordination phrase is treated as a phase (Chomsky 2001, Bošković 2020): Given that under the Phase Impenetrability Condition only the edge of a phase is accessible to operations from outside, for the filler to be able to move out of ConjP, it must be at the edge of the phase, which corresponds to the specifier of Conj and hence the first conjunct. Consequently, asymmetric extraction must take place from the first conjunct.[17]

A simplified derivation of a case-matching example with accusative (i.e, β) assigned in each conjunct like (10) is provided in the following tree diagrams (but see section 4.3 for the details regarding multiple case assignment; we use English words for ease of presentation and do not indicate possible intermediate movement steps via Spec,vP). (29) illustrates sideward movement after the second conjunct has been built:

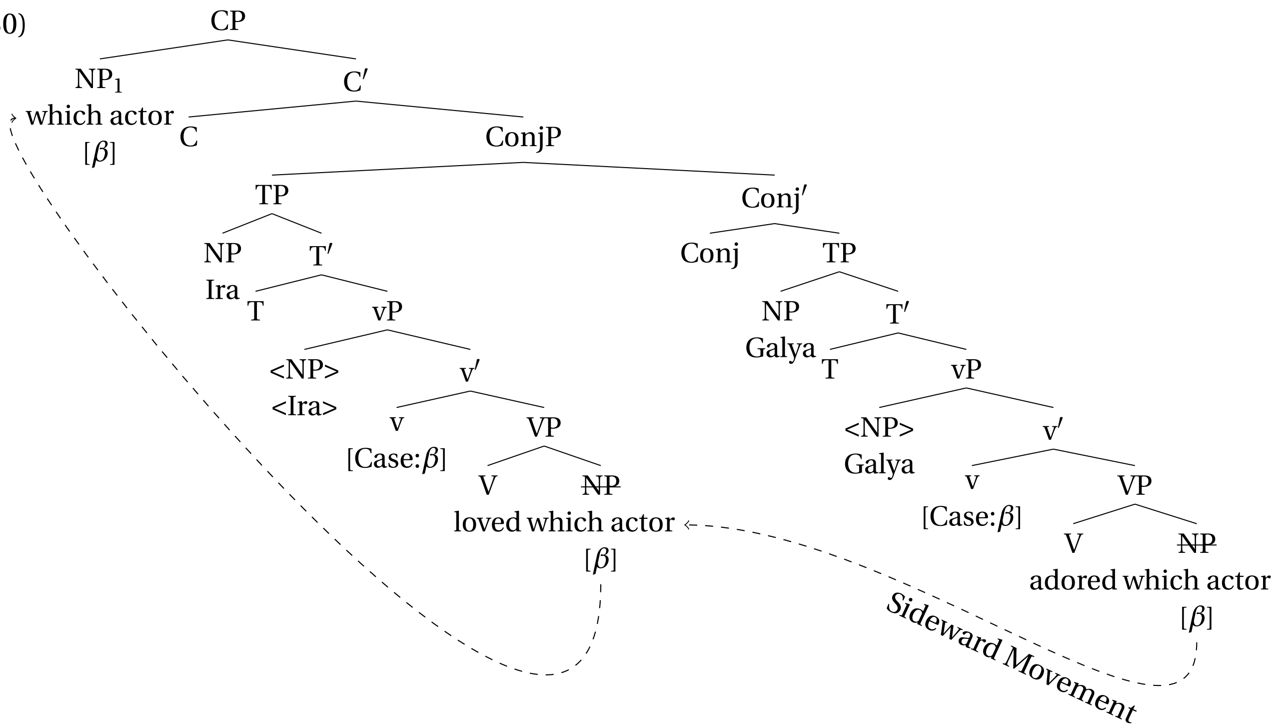
(29)



[17] The literature on ATB-movement also contains empirical arguments in favor of asymmetric extraction from the first conjunct usually related to morphological and reconstruction asymmetries (see, e.g., Munn 1993, Salzmann 2012*a*). See also Bošković (2020) for intervention-related arguments in favor of extraction from the first conjunct.

(30) illustrates Ā-movement to Spec,CP from the first conjunct after it has been built completely and merged with Conj′:

(30)



This concludes the narrow syntactic derivation of ATB-movement. Further important properties of ATB-derivations are verified at the interfaces: Regarding well-formedness at PF, depending on whether Vocabulary Insertion into the position of the filler is possible, the derivation will either converge or crash. Regarding well-formedness at LF, the derivation converges if the filler binds a variable in each conjunct (recall the discussion in 2.1.1 above).[18]

4.3 Derivations: case matching requirement

Having established the main assumptions of our model concerning case assignment and the derivation of ATB-movement in the previous section, we now show how they derive case matching in the

[18]Nunes (2004) assumes that the filler forms chains with the bottom copies in each conjunct and that chain reduction (plus possibly the elimination of unchecked formal features) produces a legible representation at the interface. Against this background, one may wonder how chain formation proceeds when the copies differ in their case-featural representation as will be the case in some of the derivations below. Note first of all that this issue arises not only with the non-syncretic mismatches we focus on here but also with syncretic mismatches quite generally, where, on most accounts, the features assigned in syntax to the copies in the first conjunct and the second, respectively, will be different. While the issue is mentioned in Nunes (2004: 176, fn. 12), he does not actually provide a solution as far as we can tell. Note that one cannot appeal to a superset condition such that higher copies must have a superset of the features of lower copies (as we assume for multiple case assignment) because in some of the syncretic mismatches, the case-features assigned in the different conjuncts are actually not in a subset-superset relationship. One thus needs to relax what counts as non-distinct regarding chain formation. Note that such a move is necessary anyway given that even in non-ATB-configurations, copies can differ in case features, in case-stacking configurations where a DP receives a second case or even in simple A-chains (subject movement to Spec,TP, ECM) where the bottom copy may have no case feature, while the topmost copy has a case value (nominative/accusative). Thus, the determination of whether two copies are distinct or non-distinct from each other will be based on their selection index (see Nunes 2004: 22f.). Still, chain formation between the filler and the copy inside the second conjunct will remain necessary to ensure that Sideward Movement is only possible if it is followed by extraction from the coordination. Otherwise, it should be possible for coordinations to simply miss an object in the second conjunct, which is normally not grammatical.

vast majority of cases, allow mismatches to be resolved by syncretism, and account for grammatical non-syncretic mismatches involving the genitive of negation. Before turning to the derivations themselves, however, we first spell out our language-specific assumptions about the position of negation and case assignment in Russian.

We follow some previous literature in assuming that the genitive is directly assigned by negation, i.e., the negative head (see also Harves 2002, Matushansky 2010,[19] Richards 2013). In simple clauses, negation always occurs on the finite verbal element, either the tensed verb or the auxiliary. We assume that NegP is located between T and vP (see also Brown and Franks 1995, Brown 1999, Harves 2002: 65, Richards 2013: 67, Gribanova 2017). To ensure correct placement, one can assume that negation reaches its surface position via additional operations such as head movement or Lowering at PF (just like finite morphology originally inserted into T, given the near-consensus that there is no verb movement to T in Russian, see Bailyn 2011: 248). Since the genitive of negation is optional in Russian, we will assume that case assignment by Neg is optional (which can be implemented in different ways, e.g., by assuming that the case feature on Neg is only optionally present).

Turning now to case assignment in Russian, we assume that the mechanisms of case assignment are established independently of the presence or absence of the genitive of negation. The genitive of negation can interact with them only indirectly, through more general constraints such as the Case Filter. In particular, we assume that accusative, as well as all other more oblique cases, are assigned in clauses with the genitive of negation in the same way as in clauses without it.

The situation with nominative is somewhat different. We assume that, in Russian, nominative assignment to unaccusative/passive subjects is in principle optional, but in the absence of an additional case nominative is nevertheless typically assigned in order to avoid a violation of the Case Filter. If another case is available, however—as in the case of the genitive of negation—nominative is not assigned. The genitive of negation will thus bleed nominative assignment in negated unaccusative and passive TPs.

We assume that the Case Filter is a clause-level constraint, i.e., it applies to each conjunct separately. This means that a case value assigned to the noun in the second conjunct, prior to its movement into the first conjunct, does not satisfy the Case Filter in the first conjunct. Nominative assignment is therefore enforced in the first conjunct, since the noun has not received case there.

Matters are again different for nominative assignment to unergative and transitive subjects, for which nominative assignment remains obligatory. We propose that arguments base-generated in Spec,vP receive an additional feature from v (e.g., an [uNom], or [uα] under our case decomposition). This feature enforces subsequent nominative assignment later in the derivation (even though nominative assignment is, in principle, optional). Naturally, if a subject is first merged in a position other than Spec,vP, it does not receive this feature. Consequently, provided that it receives some other case in the course of the derivation and hence does not violate the Case Filter, an unaccusative/passive subject need not be assigned nominative.[20] In the derivations to be discussed in this and the following (sub)sections, we will only be dealing with unaccusative subjects since only they are relevant for the present discussion (given that unergative/transitive subjects cannot bear

[19] To be precise, Matushansky (2010) actually assumes that negation assigns a q-feature.

[20] Our treatment of the distinction between VP-external and VP-internal subjects is inspired by the partitive case analysis of existential constructions in earlier literature, which makes the same cut: In this analysis, unaccusative/passive subjects (can) receive a different case, i.e., partitive, and consequently don't have to move to Spec,TP for reasons of case checking; VP-external subjects, however, can receive only nominative case, which they have to check in Spec,TP, see, e.g., Lasnik (1999) and references cited there. In our analysis, the crucial difference and consequence is not about movement for case checking to a particular position, not the least since both VP-internal and vP-external subjects can presumably remain within vP in Russian. Rather it is about whether the NP has to interact with a certain case probe. But in both approaches, the configurational difference between VP-external/VP-internal subjects affects case-marking.

genitive of negation in the first place). We return to the interaction between genitive of negation and nominative in simple clauses and the question of why only unaccusative subjects can bear genitive of negation in section 6.2.

We now begin with the full case-matching configuration, as in example (10), where structural accusative case is assigned in both conjuncts. The derivation is given in (31). It proceeds by first constructing the second conjunct: the filler is merged in the direct object position, and *v* assigns accusative case—[β] under our case decomposition. The filler is then copied into the first conjunct. Following sideward movement, it again occupies the direct object position and receives accusative/[β] from *v*.

In this derivation, the case feature assigned in the first conjunct is a superset (an improper superset, but still a superset) of the features assigned in the second conjunct, so no new feature set is created. Consequently, when the structure is transferred to morphology, the filler bears only a single feature set [β], which is faithfully realized by a fully matching accusative exponent.

(31) $[_{\text{CJ1}}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta]}\ ]\ \&\ [_{\text{CJ2}}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta]}]$

The next configuration to be discussed involves ungrammatical case mismatches. Specifically, the derivation in (32) illustrates how the analysis accounts for the data in (12-a), where a mismatch between accusative in the second conjunct and instrumental in the first conjunct leads to ungrammaticality. As in the previous derivation, *v* assigns structural accusative/[β] to the filler in the second conjunct; the filler is then copied to the first conjunct and merges as the complement of V. Assuming that lexical cases are assigned by lexical rather than functional categories, V assigns instrumental/[γ, η]—a mismatching case.

Since [γ, η] is not a superset of [β], the filler acquires a second feature set. These two feature sets are realized by different exponents: an accusative exponent is the best match for [β], while an instrumental exponent is the best match for [γ, η]. Such a derivation converges only if there exists an underspecified, syncretic exponent that can serve as the best match for both feature sets (see the assumptions in section 4.1).

In the present case, however, no such exponent exists; accusative is never syncretic with instrumental in Russian. Consequently, Vocabulary Insertion fails, and the derivation crashes at PF:

(32) $[_{\text{CJ1}}\ \text{V}_{[\gamma,\,\eta]}\ \underline{\text{wh-NP}}\ _{[\beta][\gamma,\,\eta]}\ ]\ \&\ [_{\text{CJ2}}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta]}]$

The derivation in (33) corresponds to the example in (14-a) and represents a scenario in which mismatching cases nevertheless give rise to a grammatical sentence due to the presence of a syncretic exponent. In the first step of the derivation, the noun receives accusative/[β] from *v* in the usual way. It then moves into the first conjunct, where it is assigned nominative/[α] (recall that nominative must be assigned to an NP if it hasn't received another case in the local domain). Since accusative and nominative do not stand in a superset relation, a new feature set containing the nominative feature must be created.

(33) $[_{\text{CJ1}}\ \text{T}_{[\alpha]}\ \underline{\text{wh-NP}}\ _{[\beta][\alpha]}\ ]\ \&\ [_{\text{CJ2}}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta]}]$

The noun therefore reaches morphological realization with two distinct sets of case features. Unlike in the previous derivation, however, this does not cause the derivation to crash. The noun *pis'mo* 'letter' used in (14-a) is an inanimate neuter noun of the fourth declension type and uses one and the same ending, -*o*, in both the nominative and the accusative. We assume that this ending is an elsewhere exponent and constitutes the best match for both the [α] and the [β] feature sets. It is therefore inserted, and the derivation converges.

4.4 Derivations: non-syncretic mismatches

We now present the derivations for the crucial examples involving case-mismatches with the genitive of negation. The derivation in (34) accounts for the grammatical mismatch between genitive in the first conjunct and accusative in the second conjunct (see example (17-a) above). As in the previous derivations, the filler first receives structural accusative/[β] from *v* in the second conjunct and is then copied to the first conjunct where it merges as the complement of V.

In the first conjunct, *v* assigns structural accusative case. At this stage of the derivation, the accusative/[β] feature is a superset of the existing feature set, so no new feature set is created. The derivation then proceeds with the merge of the Neg head ($\neg$), which assigns genitive—[β, γ]. Since [β, γ] is a superset of the existing feature set [β], the feature [γ] can be straightforwardly added.

This results in the feature set [β, γ] on the filler, which can be realized by a genitive exponent. The derivation thus converges.

(34) $[_{\text{CJ1}}\ \neg_{[\beta,\gamma]}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta,\gamma]}]\ \&\ [_{\text{CJ2}}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta]}]$

1 2 3✓ 4✓

Recall that the genitive-accusative mismatch leads to ungrammaticality when the order of the two conjuncts is reversed (see (17-b)). The derivation that accounts for this is provided in (35). In the second conjunct, *v* assigns structural accusative/[β] to the filler, and Neg assigns genitive/[β, γ]. Since Neg's features form a superset of the existing feature set, [γ] can be added, resulting in a single feature set [β, γ] on the filler. The filler is then copied to the first conjunct where it merges with V.

In the first conjunct, *v* assigns structural accusative/[β], which is a subset of the existing feature set. Consequently, a second feature set [β] is introduced on the filler. Although Russian in principle has exponents that are syncretic between accusative and genitive, the noun used as a filler in (17-b) belongs to a declension class that lacks such a syncretism. As a result, two different exponents are the best matches for the two feature sets (a genitive and an accusative exponent, respectively), causing Vocabulary Insertion to fail and the derivation to crash:

(35) $[_{\text{CJ1}}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta,\gamma][\beta]}]\ \&\ [_{\text{CJ2}}\ \neg_{[\beta,\gamma]}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta,\gamma]}]$

1 2✓ 3 4✗

We now turn to the mismatch involving the genitive of negation on unaccusative subjects. In one conjunct, genitive replaces nominative, while nominative is assigned to the subject in the other conjunct. This mismatch is ungrammatical regardless of the order of the conjuncts; see (19-a) for the configuration with genitive in the first conjunct and nominative in the second conjunct. The structure in (36) presents the derivation of this example.

In the second conjunct, the filler is assigned nominative/[α] by T. Recall from above that nominative is only assigned if the subject has not been assigned case yet; since there is no negation in this conjunct, there is no alternative case available for the subject/the filler. Thus, the filler is indeed caseless at this point. Consequently, it must receive nominative; otherwise, the Case Filter that applies at the conjunct-level would be violated, rendering the derivation ungrammatical.

In the next step, the filler is copied into the first conjunct where it merges with V. Here, nominative need not be assigned due to the presence of genitive of negation. Instead, the Neg head assigns genitive/$[\beta, \gamma]$ to the filler. Since Neg's feature set differs from the existing feature set $[\alpha]$, a second feature set $[\beta, \gamma]$ is introduced. In this scenario, a nominative exponent is the best match for $[\alpha]$, while a genitive exponent is the best match for $[\beta, \gamma]$. Since no single exponent is the best match for both feature sets (there are no genitive-nominative syncretisms, recall from section 4.1 above), the derivation crashes at morphological realization.[21]

(36) $[_{\text{CJ1}}\ \neg_{[\beta,\gamma]}\ \underline{\text{wh-NP}_{[\alpha][\beta,\gamma]}}]\ \&\ [_{\text{CJ2}}\ \text{T}_{[\alpha]}\ \underline{\text{wh-NP}_{[\alpha]}}]$

Finally, the derivation in (37) illustrates the analysis of the ungrammatical nominative-genitive mismatch; see the data in (19-b). In the second conjunct, given the presence of sentential negation, nominative assignment is bled; thus, Neg assigns genitive/$[\beta,\gamma]$ to the filler; the filler is then copied to the first conjunct where it merges with V.

In the first conjunct, since there are no alternative case assigners, T assigns nominative/$[\alpha]$ to the filler; recall from above that nominative must be assigned to the subject if it receives no other case *within* the clause (thus, the fact that the filler has received case in a different conjunct is irrelevant). Since T's feature set is different from the existing feature set $[\beta,\gamma]$ on the filler, a second feature set $[\alpha]$ is added to the filler. Again, the best matching exponents are different for the two feature sets, Vocabulary Insertion does not succeed and the sentence is ungrammatical.

(37) $[_{\text{CJ1}}\ \text{T}_{[\alpha]}\ \underline{\text{wh-NP}_{[\beta,\gamma][\alpha]}}]\ \&\ [_{\text{CJ2}}\ \neg_{[\beta,\gamma]}\ \underline{\text{wh-NP}_{[\beta,\gamma]}}]$

This concludes our discussion of the core examples (the derivation of the examples in (21) is identical to those in (36) and (37)). In the next section, we address additional configurations for which our proposal makes clear predictions.

5 Additional configurations

In this section, we will show that our assumptions about case decomposition and case containment make correct predictions for all additional conceivable configurations involving the three core cases nominative, accusative and genitive (lexical and genitive of negation).

We will first investigate the combination of the genitive of negation on an unaccusative subject with a direct accusative object. Then we address the combination of two genitives of negation that are based on accusative and nominative, respectively. Then, we will discuss derivations involving lexical genitives. Finally, we will address accusative fillers when combining lexical genitives with structural accusative.

[21] Note that in this example, the desired result would also obtain if nominative case were assigned in the first conjunct. Another feature $[\alpha]$ would be assigned to one of the feature sets, but that would not change the fact that two feature sets remain that cannot be realized by a single exponent. Thus, the derivation would crash as well. However, some of the derivations in section 5 crucially require the assumption that nominative be not assigned in the presence of negation, see the derivations in sections 5.1, 5.2, as well as ex. (44).

5.1 GenNeg (<Nom) & Accusative

In the first type of configuration, we combine a genitive on an unaccusative subject with structural accusative. The predictions of our approach are clear: Since the genitive contains the accusative, we expect the same pattern as with genitive on a direct object, i.e., the examples in (17): Such examples are predicted to be grammatical with genitive in the first conjunct and ungrammatical with the genitive in the second conjunct. This prediction is borne out:

(38) a. [Kakix pisem] eščë ne prišlo __, no Maša uže otpravila __?
which.PL.GEN letter.PL.GEN yet not arrived but Masha already sent
'Which letters have not arrived yet but has Masha already sent?'
GENNEG(<NOM) & ACC

b. *[Kakix pisem] Maša uže otpravila __, no eščë ne prišlo __?
which.PL.GEN letter.PL.GEN Masha already sent but yet not arrived
'Which letters has Masha already sent but haven't arrived yet?'
ACC & GENNEG(<NOM)

The derivation of (38-a) is analogous to that of GenNeg(<Acc) & Acc in (34): *v* assigns [β] to the filler in the second conjunct. In the first conjunct, ¬ assigns [β, γ] to the filler. Since this constitutes a superset, [γ] can be added to the existing feature set. Note that since the filler has already received case in the first conjunct, nominative assignment by T is bled. We thus obtain a single feature set [β, γ] that can be realized by a genitive exponent, and the derivation converges:

(39) $[_{\text{CJ1}}\ \neg_{[\beta,\gamma]}\ \underline{\text{wh-NP}}_{[\beta,\gamma]}]\ \&\ [_{\text{CJ2}}\ \text{V}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta]}]$

3✓ 2 1

The derivation of the example in (38-b) with the reversed order is analogous that of Acc & GenNeg(<Acc) in (35): The filler receives genitive, i.e., [β, γ], in the second conjunct but accusative [β] in the first conjunct. Since [β] is a subset of the existing feature set, a second feature set is created. In the absence of syncretism, there will be different best-matching exponents for the two feature sets. Vocabulary Insertion therefore fails and the derivation crashes at PF:

(40) $[_{\text{CJ1}}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta,\gamma][\beta]}]\ \&\ [_{\text{CJ2}}\ \neg_{[\beta,\gamma]}\ \underline{\text{wh-NP}}_{[\beta,\gamma]}]$

3✗ 2 1

5.2 Coordination of GenNegs

We next turn to the combination of two instances of genitive of negation where one is based on an accusative direct object and the other on an unaccusative subject. The predictions of our approach are very clear again: Since the filler receives genitive/[β, γ] in both conjuncts, the combination is predicted to be grammatical, irrespective of the order. This prediction is borne out:

(41) a. [Kakix pis'em] Katja eščë ne čitala __ i voobšče eščë ne prišlo __?
which.PL.GEN letter.PL.GEN Katja yet not read and altogether yet not arrived
'Which letters has Katja not read yet and have not arrived yet?'
GENNEG(<ACC) & GENNEG(<NOM)

b. [Kakix pis'em] eščë ne prišlo __ i Katja poetomu ne čitala __?
which.PL.GEN letter.PL.GEN yet not arrived and Katja therefore not read
'Which letters have not arrived yet and has Katja therefore not read?'
GENNEG(<NOM) & GENNEG(<ACC)

In the derivation of (41-a), the filler receives the feature set [β, γ] in the second conjunct (recall that nominative is not assigned in the presence of genitive of negation). In the first conjunct, v assigns [β]. This constitutes a subset; as a consequence, a second feature set is added to the filler. In a next step, negation assigns [β, γ] to the newly created second feature set.[22] This results in two identical feature sets [β, γ]. Since the genitive exponent is the best match for both feature sets, the derivation converges:

(42) $[_{\text{CJ1}}\ \neg_{[\beta,\gamma]}\ \text{v}_{[\beta]}\ \underline{\text{wh-NP}}_{[\beta,\gamma][\beta,\gamma]}\ ]\ \&\ [_{\text{CJ2}}\ \neg_{[\beta,\gamma]}\ \underline{\text{wh-NP}}_{[\beta,\gamma]}]$

In the derivation of (41-b) with the reversed order, the filler first receives [β] from v. Then, negation assigns [β, γ], leading to a single feature set [β, γ]. In the first conjunct, negation assigns a subset of features, i.e., [β, γ], while nominative assignment by T is bled. This results in the single feature set [β, γ] on the filler, which can be successfully realized by a genitive exponent. This derivation thus converges as well. Since this derivation is very similar to previous ones, we do not provide a schematic representation for reasons of space.

5.3 Lexical genitive

Russian has a few verbs that assign lexical genitive case, i.e., a case idiosyncratically tied to certain predicates. In this section, we will investigate how they behave in ATB-movement derivations. We will assume that the relevant verbs directly assign genitive, i.e., [β, γ].[23] Given this assumption, our analysis makes a number of predictions that we will discuss in turn.

5.3.1 Lexical genitive & GenNeg

First, we expect the lexical genitive to be compatible with genitive of negation based on accusative and nominative irrespective of the order given that they involve the same case features [β, γ]. This prediction is borne out. We first discuss derivations based on the genitive of negation with direct objects:

(43) a. [Kakix knig] Petja lišilsja __ i Katja eščë ne čitala __?
which.PL.GEN book.PL.GEN Petja lost and Katja yet not read
'Which books did Petja lose and has Katja not read yet?' LEXGEN & GENNEG(<ACC)

b. [Kakix knig] Katja eščë ne čitala __, a Petja voobšče lišilsja __?
which.PL.GEN book.PL.GEN Katja yet not read and Petja altogether lost
'Which books has Katja not read yet and did Petja even lose?' GENNEG(<ACC) & LEXGEN

In the derivation of (43-a), the filler receives the features [β] and [β,γ] from v and Neg, respectively, leading to a single feature set [β,γ] on the filler in the second conjunct. In the first conjunct, the identical feature set [β,γ] is assigned by the verb, which constitutes a superset. This results in a single feature set [β,γ] on the filler, which can be consequently realized by a genitive exponent and the derivation converges.

[22]One may wonder why the features are added to the feature set [β] and not to [β, γ]. We assume that the case features must be added to the most recently created feature set.

[23]Some verbs that assign lexical genitive contain a negative component in their lexical semantics. For verbs of this type, it is possible to assume that their lexical decomposition includes a negative head which, in parallel to the clausal negative head, is responsible for genitive assignment. While such an analysis is certainly conceivable, we leave open whether it can be generalized to the full range of lexical genitives in Russian.

In the derivation of (43-b) the filler receives the feature set $[\beta,\gamma]$ from V in the second conjunct. In the first conjunct, v assigns a subset, $[\beta]$, leading to two feature sets on the filler. The Neg head subsequently adds $[\beta,\gamma]$ to the newly created feature set. This results in two identical feature sets $[\beta,\gamma]$ on the filler, which are realized by a single genitive exponent as it is the best match for both feature sets, and the derivation converges.

The derivations that combine the lexical genitive with the genitive of negation on an unaccusative subject are next. They are also predicted to be grammatical irrespective of the order:

(44) a. Kakoj vozmožnosti Ivan lišilsja __ i u Maši tak i ne pojavilos' __?
which.GEN opportunity.GEN Ivan lost and at Masha so and not appeared
'Which opportunity did Ivan lose and never arose for Masha?'
LEXGEN & GENNEG(<NOM)

b. Kakoj vozmožnosti u Maši tak i ne pojavilos' __ i Ivan lišilsja __?
which.GEN opportunity.GEN at Masha so and not appeared and Ivan lost
'Which opportunity never arose for Masha and did Ivan lose?'
GENNEG(<NOM) & LEXGEN

In the derivation for (44-a), identical feature sets $[\beta,\gamma]$ are assigned in both conjuncts, leading to a single feature set $[\beta,\gamma]$ on the filler, which is consequently realized as genitive.

In the derivation for (44-b), identical feature sets $[\beta,\gamma]$ are again assigned in both conjuncts, leading to a single feature set $[\beta,\gamma]$ on the filler. It is realized by a genitive exponent and the derivation converges.

5.3.2 Lexical genitive & nominative/accusative

When the lexical genitive is combined with nominative or accusative, we expect the same pattern as for the combination of genitive of negation with the corresponding case.

When the lexical genitive is combined with accusative, the derivation with the lexical genitive in the first conjunct is grammatical, while the one with the lexical genitive in the second conjunct is ungrammatical:

(45) a. [Kakix prav] Petja lišilsja __ i Maša polučila __?
which.PL.GEN rights.PL.GEN Petja lost and Masha got
'Which rights did Petja lose and has Masha gained?' LEXGEN & ACC

b.*[Kakix prav] Maša polučila __ i Petja lišilsja __?
which.PL.GEN rights.PL.GEN Masha got and Petja lost
'Which rights has Masha gained and Petja lost?' ACC & LEXGEN

The derivations of (45-a) and (45-b) are parallel to (34) and (35): In the derivation for (45-a), V in the first conjunct assigns a superset $[\beta,\gamma]$ of the features assigned by v in the second conjunct ($[\beta]$). We thus obtain a single feature set $[\beta,\gamma]$ on the filler, which can be successfully realized by a genitive exponent. In the derivation for (45-b), v in the first conjunct assigns a subset $[\beta]$ of the feature set $[\beta,\gamma]$ assigned by V in the second conjunct. Consequently, a second feature set $[\beta]$ is added to the filler. Since there is no single exponent that would be the best match for both feature sets (in a declension without syncretism), Vocabulary Insertion fails and the derivation crashes.

When the lexical genitive is combined with nominative, the result is ungrammatical irrespective of the conjunct order:

(46) a.*[Kakix prav] Petja lišilsja __ i pojavilis' __ u Maši?
which.PL.GEN rights.PL.GEN Petja lost but appeared at Masha
'Which rights did Petja lose and arose for Masha?' LEXGEN & NOM

b.*[Kakix prav] pojavilis' __ u Peti i Maša lišilsja?
which.PL.GEN rights.PL.GEN appeared at Petja.DAT and Masha lost
'Which rights arose for Petja and did Masha lose?' NOM & LEXGEN

The derivations are the same as in (36) and (37), respectively.

In the derivation for (46-a) different feature sets are assigned in the two conjuncts—[α] in the second and [β,γ] in the first conjunct. Consequently, the filler obtains two feature sets. Since there is no single exponent that is the best match for both feature sets, Vocabulary insertion fails and the derivation crashes.

In the derivation for (46-b), different feature sets are again assigned in the two conjuncts—[β,γ] in the second conjunct and [α] in the first (recall that nominative must be assigned if the subject has not received case within the clause). As a result, the filler obtains two feature sets. Since there is no single exponent that is the best match for both feature sets, Vocabulary insertion fails and the derivation crashes.

5.4 Accusative filler

In this last subsection, we will address accusative fillers. In section 3.2 above we pointed out that accusative fillers are uninformative when accusative is combined with the genitive of negation because the genitive of negation is optional. Hence, irrespective of the conjunct order, we expect such configurations to be grammatical because only accusative could be assigned in each conjunct.

Things are different with the lexical genitive, which is obligatory (at least for the verbs discussed below). This allows us to test accusative fillers when structural accusative is combined with lexical genitive. The result is ungrammatical, irrespective of conjunct order:

(47) a.*[Kakie prava] Paša cenit __, a Katja lišilas' __?
which.PL.ACC right.PL.ACC Pasha values but Katja lost
'Which rights does Pasha value and did Katja lose?' Acc. filler: ACC & LEXGEN
b.*[Kakie prava] Maša lišilas' __ i Paša cenit __?
which.PL.ACC right.PL.ACC Masha lost and Pasha values
'Which rights did Katja lose and does Pasha value?' Acc. filler: LEXGEN & ACC

In the derivation for (47-a), [β,γ] is assigned by V in the second conjunct. In the first conjunct, v assigns [β], which constitutes a subset. Thus, two feature sets are created on the filler; in the absence of syncretism, the accusative exponent is not the best matching exponent for the two nor is there any other best-matching exponent for both feature sets. Hence, Vocabulary Insertion fails and the derivation crashes.

In the derivation for (47-b), [β] is assigned by *v* in the second conjunct. In the first conjunct, V assigns [β,γ]. This constitutes a superset and thus results in a single feature set [β,γ]. However, the accusative is not the best-matching exponent in this case; rather, the genitive exponent would be chosen, leading to (45-a) instead.

Thus, in other words, an example with an accusative filler cannot be generated under our approach if genitive is assigned in one of the conjuncts. Either, the derivation crashes altogether as in (47-a) or a filler with genitive obtains instead. This provides additional evidence that the contrast between (17-a) and (17-b) is not due to a proximity effect (the same holds for similar pairs like (38-a) vs. (38-b) and (45-a) vs. (45-b)).

5.5 Summary

This section concludes our discussion of additional configurations and Table 2 provides a full summary of all contexts discussed in this paper.

Let us now take stock of the data presented in this section. We began by considering contexts in which nominative-based genitive of negation is coordinated with accusative in the other conjunct, and showed that these data exhibit the same pattern as the coordination of accusative-based genitive of negation with accusative. This empirical situation follows directly from our analysis, since nominative is not assigned in the context of genitive of negation. We then turned to the coordination of two instances of genitive of negation and showed that they may be coordinated with each other in either order of the conjuncts. Next, we considered lexical genitive and showed that it behaves in exactly the same way as genitive of negation. This behavior follows from the identical feature makeup of the two cases. Finally, we turned to accusative fillers and showed that they cannot be used in coordination with genitive, since, depending on the order of the conjuncts, the derivation either crashes or must result in the filler being realized as genitive.

Table 2: Attested and unattested case configurations in Russian ATB-movement

Main pattern (section 3.2)		
GenNeg(<Acc) & Acc	OK	(17-a)
Acc & GenNeg(<Acc)	*	(17-b)
GenNeg(<Nom) & Nom	*	(19-a), (21-a)
Nom & GenNeg(<Nom)	*	(19-b), (21-b)
Nom-based GenNeg and Acc (section 5.1)		
GenNeg(<Nom) & Acc	OK	(38-a)
Acc & GenNeg(<Nom)	*	(38-b)
Coordination of GenNegs (section 5.2)		
GenNeg(<Acc) & GenNeg(<Nom)	OK	(41-a)
GenNeg(<Nom) & GenNeg(<Acc)	OK	(41-b)
LexGen and GenNeg (section 5.3.1)		
LexGen & GenNeg(<Acc)	OK	(43-a)
GenNeg(<Acc) & LexGen	OK	(43-b)
LexGen & GenNeg(<Nom)	OK	(44-a)
GenNeg(<Nom) & LexGen	OK	(44-b)
LexGen and Acc/Nom (section 5.3.2)		
LexGen & Acc	OK	(45-a)
Acc & LexGen	*	(45-b)
LexGen & Nom	*	(46-a)
Nom & LexGen	*	(46-b)
Accusative filler (section 5.4)		
Acc filler – Acc & LexGen	*	(47-a)
Acc filler – LexGen & Acc	*	(47-b)

6 The distribution and the analysis of the genitive of negation

The concept of multiple case assignment and the decomposition of case features are at the core of the analysis developed for grammatical non-syncretic mismatches in ATB-movement. In this section, we show that these mechanisms also allow us to adequately restrict the distribution of the genitive of negation in a simple clause.

6.1 The distribution of the genitive negation

We begin by going through the contexts where genitive of negation is grammatical in Russian. By going through different contexts reported in the previous literature and new ones, we can formulate the following new generalization: The genitive of negation is very broadly grammatical on nouns that are in the c-command domain of negation and would be marked accusative otherwise.

First, genitive of negation is not restricted to complements of V (as has been claimed in previous literature), but can also appear on an otherwise accusative argument if it is the higher one and presumably occupies the Spec,VP position. This is illustrated in (48)-(49) for higher accusative objects followed by a PP.

(48) Učitel' ne položil svoi knigi / svoix knig na stol.
teacher not put self.PL.ACC book.PL.ACC self.PL.GEN book.PL.GEN on table.ACC
'The teacher has not put his books on the table.'

(49) Étot sudja ne obvinil i odnu ženščiny / i odnoj ženščiny vo vran'e.
this judge not accused even one.ACC woman.ACC even one.GEN woman.GEN in lying
'This judge did not accuse a single woman of lying.'

Second, example (50) illustrates that genitive of negation is grammatical with ditransitive accusative-dative verbs for which it was argued that the accusative argument is the higher one (Boneh and Nash 2017).

(50) On ne upodobil čuvstva / čuvstv razumu.
he not likened feeling.PL.ACC feeling.PL.GEN mind.DAT
'He did not liken feelings to reason.'

Third, genitive of negation can appear on raised objects in raising-to-object contexts as in (51). Raising to object in Russian is restricted to adjectival predicates (Lyutikova 2022*a*,*b*).

(51) Ona ne sčitaet i odnu devočku / i odnoj devočki glupoj.
she not consider even one.ACC girl.ACC even one.GEN girl.GEN stupid.INS
'She does not consider a single girl stupid.'

Fourth, genitive of negation is grammatical on accusative objects independently of their Θ-role (pace earlier claims by Legendre and Akimova 1994). This is illustrated by the data in (52), where the accusative object is an experiencer.

(52) Étot šum ne pobespokoil našu sosedku / našej sosedki.
this noise not bothered our.ACC neighbor.ACC our.GEN neighbor.GEN
'This noise did not bother our neighbor.'

Fifth, genitive of negation is not restricted to arguments; it can appear on otherwise accusative adverbials:

(53) On ne čital ètoj knigi i dve minuty / i dvux minut.
he not read this book.GEN even two.ACC minute.ACC even two.GEN minute.GEN
'He did not even read this book for two minutes.'

This use of the genitive of negation has been somewhat controversial in the previous literature. Franks and Dziwirek (1993) have argued that it differs from other instances of the genitive of negation and therefore should not be analyzed as a genitive of negation but rather as a partitive. Nevertheless, we continue treating this as an instance of genitive of negation because unlike partitive genitive, the genitive on adverbials is only possible in the context of negation (see Borovikoff 1997).

The presence of semantic effects is also not specific to genitive on adverbials, but is attested for all cases of genitive of negation to various degrees, as mentioned in section 2.2.

Sixth, genitive of negation can apply long-distance in that it affects accusative arguments of an embedded (subject-control) infinitive (Harves 2002: 72); see (54).

(54) Ja ne xoču videt' ètu knigu / ètoj knigi.
I not want see this.ACC book.ACC this.GEN book.GEN
'I do not want to see this book.'

Note however that genitive of negation is constrained by the established locality domains in Russian: Genitive assignment cannot apply across finite CP-boundaries as in (55-a), and it also cannot reach into object control infinitives, which have been argued to contain more structure than subject control infinitives, (55-b) (Neidle 1988, Franks 1995: 199, Harves 2002: 72):

(55) a. On ne skazal, čto Katja videla etu knigu / *étoj knigi.
he not said that Katja saw this.ACC book.ACC this.GEN book.GEN
'He did not say that Katja saw this book.'
b. Učitel' ne vynušdal Mašu čitat' étu knigu / *étoj knigi.
teacher not forced Masha.ACC read this.ACC book.ACC this.GEN book.GEN
'The teacher did not force Masha to read this book.'

Genitive of negation also cannot cross PP-layers: it does not appear on complements of prepositions assigning accusative case. Under the assumption that PPs are phases, only accusative case is assigned to such NPs; the genitive assigned by negation is thus blocked by the PP.

(56) a. Ira pošla v školu.
Ira went in school.ACC
'Ira went to school.'
b. Ira ne pošla v školu / *v školy.
Ira not went in school.ACC in school.GEN
'Ira did not go to school.'

Finally, the only case where genitive of negation is largely degraded on a local direct object comes from sentences with object control; see (57).

(57) Roduteli ne naučili i odnu devočku / *i odnoj devočki pravil'no padat'.
parents not taught even one.ACC girl.ACC even one.GEN girl.GEN correctly fall
'Parents did not teach even a single girl to fall correctly.'

We assume that the ungrammaticality in this case is not due to the position of the noun phrase, but rather due to the limited ability of an argument with genitive of negation to control PRO in the infinitival clause (see Neidle 1988, Babyonyshev 1997).

Turning now to nominative arguments, the generalization proposed in the previous literature draws a distinction between unaccusative and passive subjects, which allow the genitive of negation, and unergative and transitive subjects, which do not:

(58) a. Devočka ne igrala.
girl.NOM not play.PST.F.SG
b.*Devočki ne igralo.
girl.GEN not playd.PST.N.SG
'A girl did not play.'

(59) a.*Devočka ne čitaet knigu.
girl.NOM not read book
b.*Devočki ne čitaet knigu.
girl.GEN not read book
'A girl does not read a book.'

The data also show that the genitive of negation is ungrammatical on the controller in cases of subject control. This is likely due both to the ungrammaticality of control itself (i.e., the inability of genitive to control PRO) and to the structural position of the argument in the matrix clause (i.e., the fact that it is an external argument).

(60) a. Rebjata ne xoteli pojti v bar.
guys.PL.NOM not want.PST.PL go to bar
b.*Rebjat ne xotelo pojti v bar.
guys.PL.GEN not want.PST.N.SG go to bar
'The guys didn't want to go to the bar.'

More interesting data obtain with raising: although judgments are not completely uniform across speakers, it appears that the acceptability of the genitive of negation on raised subjects depends on the base position of the subject in the embedded clause. The genitive of negation is considerably more acceptable with a raised unaccusative argument (see (61)) than with an argument that prior to raising occupies the position of a transitive subject in the embedded clause (see (62)).

(61) a. Na étoj ploščadke i odna devočka ne mogla upast'.
on this playground even one.NOM girl.NOM not can.PST.F.SG fall
b.?Na étoj ploščadke i odnoj devočki ne moglo upast'.
on this playground even one.GEN girl.GEN not can.PST.N.SG fall
'Not even a single girl could fall on this playground.'

(62) a. V étom detskom sadu i odna devočka ne mogla est' kašu.
in this children garden even one.NOM girl.NOM not can.PST.F.SG eat porridge
b.*V étom detskom sadu i odnoj devočki ne moglo est' kašu.
in this children garden even one.GEN girl.GEN not can.PST.N.SG eat porridge
'Not even a single girl could eat porridge in this daycare.'

To complete the picture, the genitive of negation never appears on arguments that would otherwise bear a case other than accusative or nominative. Example (63-a) illustrates this for dative and (63-b) for instrumental.

(63) a. Ona ne pomogla Ivanu / *Ivana.
she not helped Ivan.DAT Ivan.GEN
'She did not help Ivan.'
b. Ona ne pol'zovalas' komp'juterom / *komp'jutera.
she not used computer.INS computer.GEN
'She did not use a computer.'

Summarizing this new data, we propose that the correct generalization governing the distribution of the genitive of negation is the following: the genitive of negation is possible on accusative noun phrases that are c-commanded by negation and are sufficiently local to it, and on nominative arguments, provided that they are base-generated in a position other than Spec,vP.

6.2 New approach to genitive of negation

We now go through the derivations of genitive of negation in different contexts and show that the assumptions regarding multiple case assignment and case decomposition that are required to account for limited non-syncretic mismatches in ATB-movement also allow us to derive the distribution of genitive of negation in simple clauses.

In all cases where the genitive appears on a noun that would otherwise be marked accusative, genitive case can be straightforwardly added on top of accusative. This is because accusative/[β] is fully contained within genitive/[β,γ]. In (64), a derivation of the genitive of negation is shown for the accusative complement of V as in, e.g, example (7-b); but nothing in the derivation refers to this particular structural position, and the same mechanism applies equally to any accusative-marked NP that is c-commanded by negation. Accusative is assigned in the first step, and genitive is subsequently added to the same feature set, since genitive is a superset of accusative. The noun is then faithfully realized by a genitive exponent in morphology.

(64) [T [$\neg_{[\beta,\gamma]}$ [NP $\text{v}_{[\beta]}$ [V $\text{NP}_{[\beta,\gamma]}$]]]] $^{\text{OK}}$Gen + Acc

2✓ 1

The proposed mechanism of multiple case assignment predicts that if an NP has already been assigned a case other than accusative earlier in the derivation, the addition of genitive will result in the creation of a second case feature set. This, in turn, will lead to ungrammaticality in the absence of a syncretic exponent.

The derivation in (65) accounts for the ungrammaticality of genitive on an argument that would otherwise be marked dative, as, e.g., in (63-a). The noun receives dative/[γ,δ,ϵ] in the first step of case assignment; genitive/[β,γ] assigned by negation later overlaps with the features of the dative, but does not fully contain the dative's feature set. The noun phrase hence receives a second feature set and the derivation crashes at spell-out (in the absence of a syncretic exponent).

(65) [T [$\neg_{[\beta,\gamma]}$ [NP v [$\text{V}_{[\gamma,\delta,\epsilon]}$ $\text{NP}_{[\gamma,\delta,\epsilon][\beta,\gamma]}$]]]] *Gen + Dat

2✗ 1

Essentially the same situation arises when the genitive of negation is assigned to an argument that receives nominative. The only difference lies in the order of case assignment: nominative is assigned by T, which, under our assumptions, is located higher than the negation head responsible for assigning genitive (recall from section 4.3 above).

We begin with unaccusative/passive subjects. As proposed in section 4.1, we assume that nominative assignment to such subjects by T is not obligatory; it is assigned if the subject has not received a case within the clause. As a consequence, nominative is in fact not assigned to unaccusative and passive subjects already bearing genitive. As a result, genitive is the only case assigned to these NPs, and there is therefore no reason for the derivation to crash. The subject is faithfully realized by a genitive exponent at PF, see (66) (the derivation is thus identical to what happens in the second conjunct of ATB-derivations with genitive of negation assigned to an unaccusative subject, as, e.g., in (37), (38-b), (43) above):

(66) [T [$\neg_{[\beta,\gamma]}$ [v [V $\text{NP}_{[\beta,\gamma]}$]]]] Gen only

1

Turning to unergative/transitive subjects, recall from section 4.1 that subjects projected in Spec,vP receive a [uα]-feature from v. This feature forces the subject to get nominative case from T. As a consequence, as shown in (67), upon receiving Gen/[β,γ] from Neg, the subject also receives nominative/[α] from T. Since [α] is not a superset of [β,γ], a second feature set is created on the filler. Since there are no exponents syncretic for genitive-nominative in Russian, the derivation crashes at Vocabulary Insertion:

(67) $[\, \mathrm{T}_{[\alpha]}\ [\, \neg_{[\beta,\gamma]}\ [\, \mathrm{NP}_{[\alpha][\beta,\gamma]}\ \mathrm{v}\ [\, \mathrm{V}\ \mathrm{NP}\,]\,]\,]\,]$ *Gen + Nom

The crucial difference between unaccusative and unergative/transitive subjects thus lies in the (non-)optionality of nominative case assignment. Given that nominative and genitive are not in a subset-superset relationship, a derivation where the subject receives both cases is doomed to crash.

In the last part of this section, we will briefly discuss previous approaches to the genitive of negation, both in relation to its distribution in simple clauses as well as regarding its behavior in ATB-movement.

6.3 Previous approaches to the genitive of negation and ATB-movement

We have shown so far that our approach based on multiple case assignment and partial case containment achieves two things: It derives why genitive is special regarding case-mismatches in ATB-movement and at the same time accounts for its distribution in single clauses with the same means.

Previous approaches to the genitive of negation fail to account for one or both of these components. Regarding the distribution in a single clause, Pesetsky (1982), Harves (2002) and Bailyn (2011) link the genitive of negation to the complement position of the verb; but as shown in section 6.1, the genitive of negation can also occur on arguments of V that are not complements (but specifiers of V or specifiers of some functional head above V). Thus, these accounts undergenerate. Furthermore, these accounts are designed to account for the genitive of negation in simple clauses, but they cannot obviously be extended to deal with multiple case assignment in ATB-movement, while our approach relies on multiple case assignment, which offers the possibility to deal with both the genitive of negation in simple clauses and case-matching in ATB-movement.

Regarding case-mismatches in ATB-movement, relevant predictions only arise in the approaches by Matushansky (2010) and Sigurðsson and Šereikaitė (2024): Matushansky (2010) is the only one that explicitly foresees the possibility of multiple case assignment, and the PF-approach by Sigurðsson and Šereikaitė (2024) may provide a handle on the non-syncretic mismatches. In Matushansky (2010), cases are assigned to the case-assigner's sister and percolate down to any NP c-commanded by the case assigner. NPs can thus be assigned multiple cases. The choice of exponent then depends on its specificity. For the correct result to obtain, lexical cases are thus more specific than structural cases. This motivates why secondary non-verbal predicates bear a more oblique case than, say, subjects and direct object. Similarly, since genitive is more specific than nominative and accusative, a subject bearing nominative and genitive or a direct object bearing accusative and genitive will surface as genitive. Lexical cases, on the other hand, are more specific than genitive, which is why they are not overwritten by the genitive of negation. At any rate, the expectation for ATB-movement is very clear: Since a DP can receive multiple cases without there being constraints on what can be combined, the prediction is that any case-mismatch should be possible: whichever case is more oblique (= more specific) in the two conjuncts should surface on the filler, irrespective of the order. This clearly overgenerates in that it wrongly predicts, e.g., (12-a) and (13-a) to be grammatical.

In Sigurðsson and Šereikaitė (2024), the genitive of negation is due to a PF-rule that changes the case assigned in syntax to genitive before Vocabulary Insertion. Extending their approach to Russian,[24] any (copy of a) DP bearing accusative or nominative is changed to genitive if c-

[24] To be precise, in their analysis of the genitive of negation in Lithuanian, Sigurðsson and Šereikaitė (2024) restrict this rule to accusative since nominative is not affected by the genitive of negation in the language. We thus evaluate a modified version of their approach to be able to apply it to Russian and hence add a rule for nominative to genitive

commanded by sentential negation. This accounts for the distribution of the genitive of negation in a simple clause in that it only targets DPs marked accusative or nominative. Assuming that case-matching is (at least in part) verified in syntax, the PF-perspective also provides a handle on the genitive-accusative mismatch in (17-a): Since syntactically, only accusative is assigned, there is case-matching and the sentence is hence predicted to be grammatical; one will need to assume that the fact that at PF, the copy of the filler in the first conjunct is changed to genitive does not matter for case matching (see Bondaruk 2003: 232f. for a similar suggestion). However, the approach both over- and undergenerates in a number of configurations, having to do with the fact that syntactically, there is no genitive. Concretely, this leads to an incorrect prediction for the derivation of (19-a) where genitive of negation on a subject is combined with nominative in the second conjunct: Since the filler receives nominative both in the second and the first conjunct, the case-matching requirement should be satisfied. As with the genitive-accusative mismatch in (17-a), the fact that the copy in the first conjunct is realized as genitive at PF should not negatively impact case-matching. Thus, (19-a) is incorrectly predicted to be grammatical.[25] Further incorrect predictions arise in the combination of genitive of negation on a nominative with accusative in the second conjunct, see (38-a), which would constitute a nominative-accusative mismatch in syntax under the PF-approach and thus should be ungrammatical, contrary to fact. The same kind of nominative-accusative mismatch would obtain in the combination of genitive of negation on a nominative with genitive of negation on a direct object (irrespective of order) as in (41); these examples are thus also incorrectly predicted to be ungrammatical. Similarly, in the combination of a lexical genitive with the genitive of negation based on accusative or nominative as in (43) and (44), one would be dealing with a genitive-nominative/accusative mismatch in syntax, which hence should also lead to ungrammaticality.

Thus, to conclude this section, unlike previous accounts of the genitive of negation, our proposal not only captures limited non-syncretic mismatches in ATB-movement, it simultaneously accounts for the distribution of the genitive of negation in simple clauses.

7 Theoretical implications for the syntax of ATB-movement

In our analysis above, we have adopted a sideward movement approach to ATB-movement. We argue that this approach has exactly the right properties to capture the aspects of ATB-movement in Russian relevant to our inquiry: It can account for case matching and deal with both syncretic and (grammatical) non-syncretic mismatches. We will show in what follows that alternative approaches to ATB-movement fail to account for at least one of these properties.

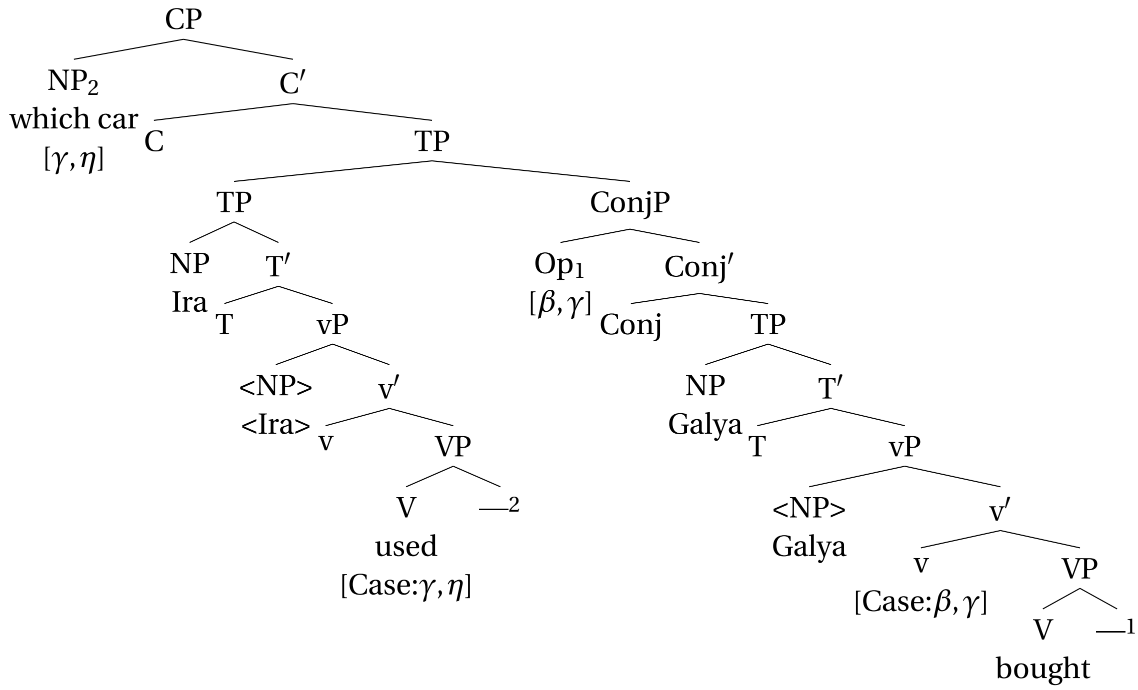
With empty operator (Munn 1993, Franks 1993, 1995) and ellipsis (Ha 2008, Salzmann 2012*a*) approaches it is not clear in the first place whether the case matching requirement can be implemented. Fusion (Hein and Murphy 2020) and multidominance approaches (Citko 2005, 2011, Bachrach and Kazir 2009, Gračanin-Yuksek 2013), on the other hand, are designed to capture case-matching but cannot account for grammatical non-syncretic mismatches.

In empty operator/parasitic gap-type approaches there are usually two independent chains with a null operator moving in the second conjunct, see (68) for an example combining instrumental and accusative case as in ex. (12-a) (we assume coordination at the TP-level, which involves adjunction of a Boolean phrase in Munn (1993), but we represent it as ConjP and use English words for ease of representation):

conversion at PF.

[25]Things are different with (19-b), where the filler would surface with nominative (since negation is in the second conjunct and could not affect the copy in the first conjunct).

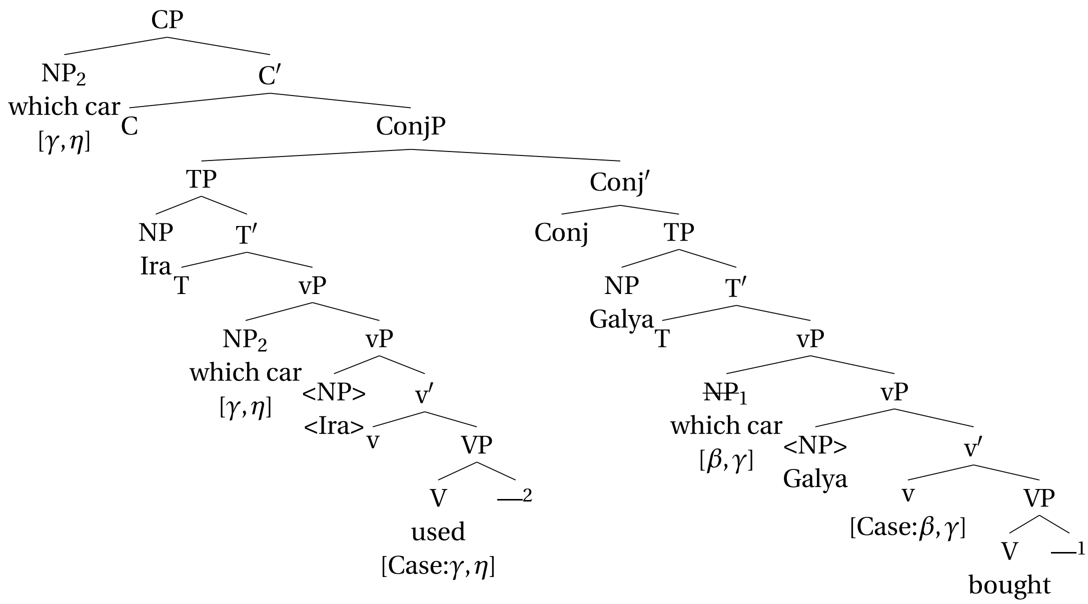
(68)



Some operation of chain composition is then needed to ensure that only the overt filler is interpreted and linked to both conjuncts. The fact that parasitic gaps are subject to the same case-matching restrictions as ATB-movement, including the alleviating effect of syncretism (Franks 1993, 1995, Bondaruk 2003, but see Hein and Murphy 2020: 263 for counterexamples) may initially provide support for such an approach, but nothing in the grammar of operator movement/parasitic gaps actually enforces case matching. Rather, this would have to be added to whatever mechanism composes the two chains and hence wouldn't follow from the approach as such (see Franks 1995: 80f.). In addition, there remain systematic asymmetries between the two constructions that complicate a unification.

In ellipsis approaches, an instance of the filler is merged in each conjunct and undergoes movement; before merger with Conj, the filler in one of the conjuncts is deleted under identity with a copy of the filler in the other conjunct. This is illustrated in (69) for an example combining instrumental and accusative case as in example (12-a) (we assume forward ellipsis with movement up to Spec,vP in the second conjunct and again use English words for ease of representation):

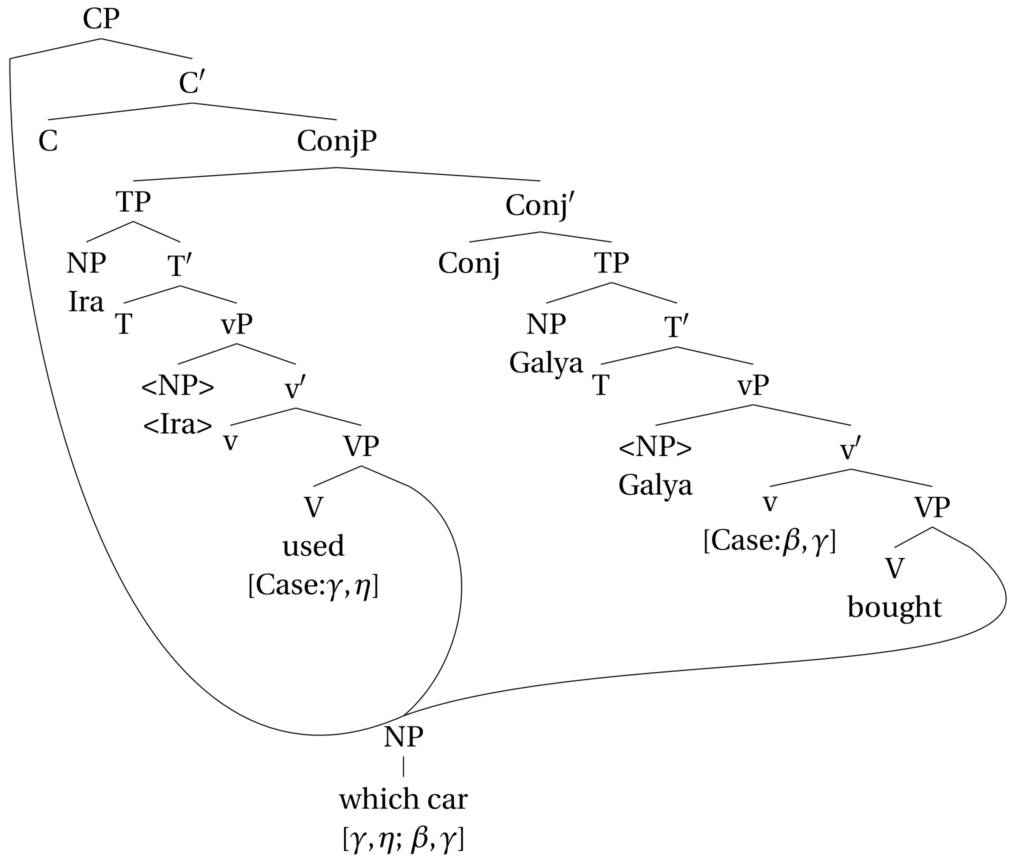
(69)



As discussed in Hein and Murphy (2020: 266f.), while ellipsis tolerates mismatches, the profile of the tolerated mismatches is different from the case-mismatches possible in ATB-movement: It tolerates mismatches in form as long as the features match, while in ATB-movement, mismatches in case features are tolerated, while the form must match (i.e. syncretism). Thus, it is not clear how case matching could be enforced and how the syncretism effect could be derived; rather, one may expect case-mismatches to be systematically possible.

In multidominance approaches, a single filler is linked to a position in each conjunct and Spec,CP. This is illustrated in (70) for an example combining instrumental and accusative case as in example (12-a):

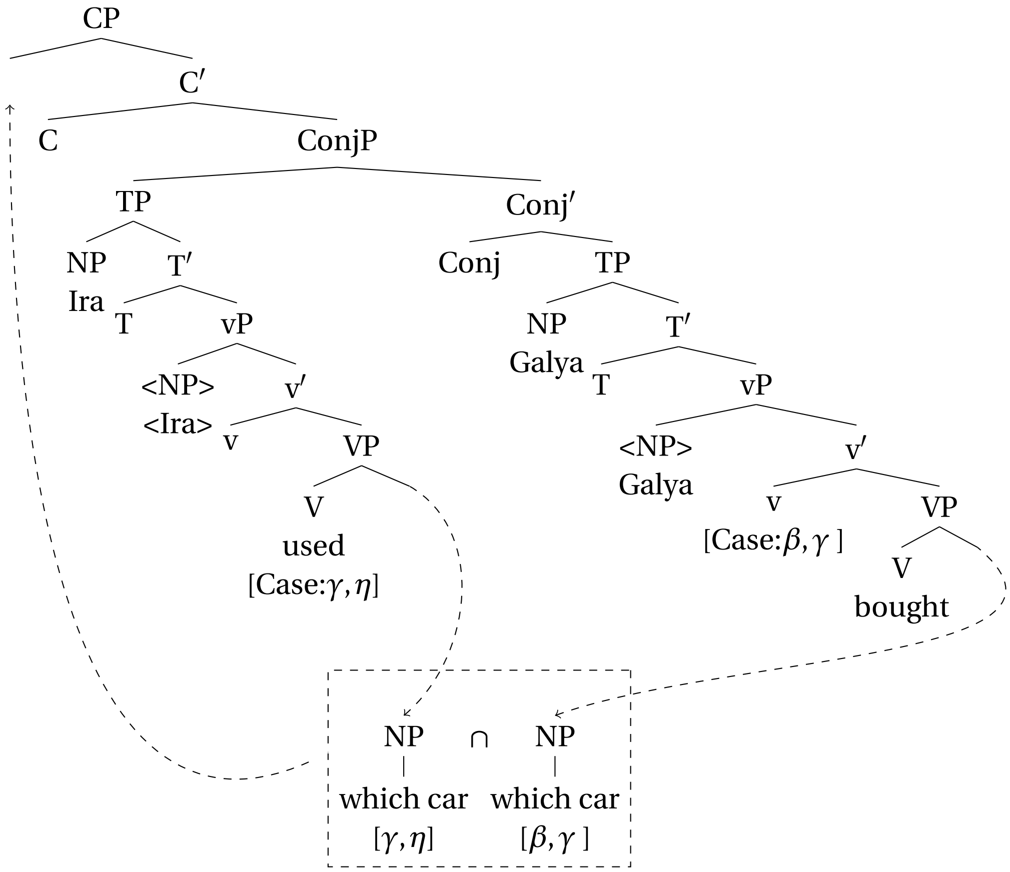
(70)



Citko (2005) intends to account for case-matching and the syncretism effect by assuming that the filler receives two case features, which can be realized only if there is a form that is compatible with both features (an underspecified form in case of syncretism). However, as shown in Hein and Murphy (2020: 270-272), this account actually fails to rule out case-mismatches: given the subset principle, inserting a marker that only matches one of the case features should be possible, contrary to fact. One could, of course, amend the approach with our account of syncretism and the same rules for Vocabulary Insertion, i.e., one posits two feature sets on the filler and assumes that a syncretic exponent has to be the best match for both feature sets for insertion to be possible. This would correctly rule out ungrammatical non-syncretic case-mismatches and rule in (grammatical) syncretic case-mismatches. However, it would still fail to account for the fact that only certain non-syncretic mismatches involving the genitive of negation (i.e., (17-a) but not (17-b)) are grammatical. The crucial point seems to be that in this kind of framework, one cannot easily make a derivational difference between case assignment in the second conjunct and case assignment in the first: Parallel merge of the filler with the verb in the respective conjuncts is simultaneous and the conjuncts are then built incrementally in parallel. What then matters are differences regarding the height of the probes inside the conjuncts. In our crucial examples combining accusative with the genitive of negation (and equivalent examples discussed in section 5), accusative would be assigned simultaneously to the single filler given that it is assigned by the same head, i.e., v (see Citko 2005: 481). Genitive would be assigned next, but this should lead to a superset irrespective of the order of conjuncts given that the only derivational difference would be the height of the case probes but not which conjunct they are in. Consequently, both (17-a) and (17-b) would be predicted to be grammatical, contrary to fact.

In the fusion approach by Hein and Murphy (2020) there is parallel extraction from both conjuncts. The two extracted fillers move to an external workspace where they are fused by means of intersection of their feature sets. The fused syntactic object is then merged in the landing site. This is illustrated in (71) for an example combining instrumental and accusative case as in example (12-a):

(71)



Cases are decomposed into [±sub, ±gov, ±obl] and ATB-movement converges if, after set intersection, there is an exponent that can be inserted into the position of the filler in Spec,CP. The syncretism effect obtains if after set intersection, an underspecified exponent can be inserted. Non-syncretic case mismatches are ruled out because either (i) the resulting feature set is incompatible with any of the available vocabulary items or (ii) the resulting feature set is empty, violating the Case Filter/Full Interpretation. The system is constructed such that syncretic case-mismatches are grammatical, while non-syncretic mismatches are ruled out. It would thus correctly rule out examples like (12) and (13), but it would also incorrectly rule out the grammatical non-syncretic mismatches involving the genitive of negation, i.e., (17-a) (and equivalent examples in section 5). To rule in (17-a), one could try to combine fusion with our decomposition; however, non-syncretic mismatches with GenNeg(<Acc) & Acc and Acc & GenNeg(<Acc) would converge with Acc on the filler rather than genitive, due to feature set intersection ($[\beta] \cap [\beta,\gamma] = [\beta]$). While, as discussed above, the ungrammaticality of accusative fillers in such configurations cannot easily be shown with the genitive of negation, the examples with the lexical genitive discussed in section 5.4 clearly

show that an accusative filler is ungrammatical if genitive is assigned in any of the conjuncts and accusative in the other. Thus, while in our sideward movement approach the number of case features can grow in the course of the derivation, it is invariably reduced under the fusion approach. It consequently cannot account for the fact that in the (17-a) configuration, the filler ends up bearing both genitive and accusative case features. Finally, any conceivable additional modifications would not be able to capture the relevance of the position of negation and hence the difference between (17-a) and (17-b)—parallel extraction takes place simultaneously and thus there is no derivational ordering of operations that could affect the interaction of cases.

To summarize, all alternative approaches to ATB-movement fail to account for certain aspects of case-matching in ATB-movement: They either do not impose any restrictions on case matching (empty operator approach, ellipsis approach) or predict only syncretic case-mismatches to be possible (multidominance, fusion approach). The sideward movement approach as implemented here is clearly at an advantage in that it rules out ungrammatical non-syncretic mismatches and captures both syncretic mismatches and grammatical non-syncretic mismatches. This is partially due to our assumptions about case decomposition and case containment, and partially due to the fact that in our sideward movement approach, the timing of case assignment (second conjunct before first conjunct) matters.

8 Conclusion

In this paper, we have presented the first detailed description and analysis of grammatical non-syncretic mismatches in Russian ATB-movement. While Russian patterns like many other languages in that ATB-movement is generally subject to a case-matching requirement (modulo mismatches resolved by syncretism), limited non-syncretic mismatches are possible when the genitive of negation is involved. We have shown that such mismatches are possible if accusative-based genitive of negation appears in the first conjunct and accusative is assigned to the filler in the second conjunct. The mismatch is ungrammatical if the order of conjuncts is reversed and it is also ungrammatical if genitive of negation replaces nominative and nominative is used in the other conjunct.

To account for both the case-matching configurations as well as examples with non-syncretic mismatches, we have proposed an account based on multiple case assignment and partial case containment. Regarding decomposition and containment, we have assumed that the genitive featurally subsumes the accusative but not the nominative. General principles of multiple case assignment then determine whether certain cases can be combined with each other. Concretely, if the case features assigned in the first conjunct are a superset of those assigned in the second conjunct, any additional features can be added to the existing feature set, leading to a single feature set that can be realized by an appropriate exponent. If the case features assigned in the first conjunct are not a superset of those assigned in the second conjunct, a second feature set is created on the filler. If there is no exponent that is the best match for both feature sets, the derivation crashes at Vocabulary Insertion. If there is a syncretic marker that is the best match for both feature sets, the derivation can converge.

We have further reviewed the distribution of genitive of negation in simple clauses. We have shown that our assumptions about multiple case assignment and case containment also correctly restrict its distribution in that domain. Our approach thus achieves two things: It not only explains why genitive is special regarding case-mismatches in ATB-movement, it also accounts for its distribution in single clauses with the same means.

In the last part of the paper, we have explored the consequences of our new data for theories of ATB-movement. We have shown that most existing approaches are ill-suited to deal with both case-matching and limited non-syncretic mismatches at the same time: Some asymmetric approaches (operator movement, ellipsis) fail to impose any limits on case-matching and hence overgenerate. Symmetric approaches are usually explicitly designed to allow only syncretic mismatches (multidominance, parallel extraction) and hence undergenerate. Even if such approaches were amended with our assumptions about decomposition/containment and multiple case assignment, they would still not achieve the same empirical coverage: Under multidominance, there is no machinery to differentiate between the case assigned in the first and in the second conjunct. As a result, it incorrectly predicts both the grammatical genitive-accusative mismatch in (17-a) and the ungrammatical accusative-genitive mismatch in (17-b) to be well-formed. Under parallel extraction with set intersection, both configurations are also predicted to be grammatical, but with the filler bearing accusative, a possibility we show to be unavailable in section 5.4). We have instead proposed on account in terms of sideward movement. This approach has exactly the properties needed to account for the data in this paper: First, due to copying from one conjunct to the other, it generally provides a handle on implementing case-matching as the cases are assigned to the same filler and rules regarding the interaction of the cases can be formulated. Secondly, given that the filler is copied from the second conjunct to the first, the interaction of the cases can be sensitive to the timing of the cases assigned (unlike in the symmetrical approaches discussed above).

Before concluding this paper, we would like to briefly discuss what makes the genitive special in Russian—regarding its featural composition, the containment of accusative but not nominative and hence its behavior in ATB-movement. Obviously, these are properties that set it apart from other cases in Russian and from genitives in other languages. While we believe that this is, to some extent, simply a fact about the language, a more general explanation suggests itself once it is recognized that the genitive of negation is an instance of differential object marking—it alternates with a different case, accusative, on the same argument. While languages may of course differ in how differential case marking is formally implemented, we tentatively suggest that one possible source of alternations between two cases lies in their featural similarity and in the corresponding containment relation between them. Although such an approach is unlikely to extend to all languages, it predicts that non-syncretic mismatches could be available in other languages with differential object marking as well. We leave the investigation of these possible predictions for future work.

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